



**RV College
of
Engineering**

*Go, change the
world*

Unit 4– Traditional Block Cipher and Public key Cryptosystem

Contents

- Stream Ciphers and Block Ciphers, Feistel Cipher Structure.
- The Data Encryption Standard-Encryption and Decryption.
- Principles of Public Cryptosystems- Public-Key Cryptosystems.
- Applications for Public-Key Cryptosystems.
- Requirements for Public-Key Cryptosystems.
- Public-Key Cryptanalysis.
- The RSA algorithm-Algorithm and Computational Aspects and the security of RSA.
- Other Public key Cryptosystems: Diffie-Hellman Key Exchange.

Introduction

- Focus on two broad areas:
 1. Cryptographic algorithms and protocols
 - Symmetric Encryption.
 - Asymmetric Encryption.
 - Data Integrity algorithms
 - Authentication protocols
 2. Network and Internet security: consists of measures to deter, prevent, detect, and correct security violations that involve the transmission of information.

Symmetric Encryption: Used to conceal the contents of blocks or streams of data of any size, including messages, files, encryption keys, and passwords.

- Asymmetric Encryption: Used to conceal small blocks of data, such as encryption keys and hash function values, which are used in digital signatures.
- Data Integrity algorithms: Used to protect blocks of data, such as messages, from alteration.
- Authentication protocols: These are schemes based on the use of cryptographic algorithms designed to authenticate the identity of entities.

Examples of Security violations:

1. User A transmits a file to user B. The file contains sensitive information (e.g., payroll records) that is to be protected from disclosure. User C, who is not authorized to read the file, is able to monitor the transmission and capture a copy of the file during its transmission.

Examples of Security violations:

2. A network manager, D, transmits a message to a computer, E, under its management. The message instructs computer E to update an authorization file to include the identities of a number of new users who are to be given access to that computer. User F intercepts the message, alters its contents to add or delete entries, and then forwards the message to computer E, which accepts the message as coming from manager D and updates its authorization file accordingly

Examples of Security violations:

3. Rather than intercept a message, user F constructs its own message with the desired entries and transmits that message to computer E as if it had come from manager D. Computer E accepts the message as coming from manager D and updates its authorization file accordingly

Examples of Security violations:

4. An employee is fired without warning. The personnel manager sends a message to a server system to invalidate the employee's account. When the invalidation is accomplished, the server is to post a notice to the employee's file as confirmation of the action. The employee is able to intercept the message and delay it long enough to make a final access to the server to retrieve sensitive information. The message is then forwarded, the action taken, and the confirmation posted. The employee's action may go unnoticed for some considerable time

Examples of Security violations:

5. A message is sent from a customer to a stockbroker with instructions for various transactions. Subsequently, the investments lose value and the customer denies sending the message.

Definition of Computer Security:

- The NIST(National Institute of Standards and Technology) Computer

Security Handbook [NIST95] defines the term computer security as follows:

“The protection afforded to an automated information system in order to attain the applicable objectives of preserving the integrity, availability, and confidentiality of information system resources (includes hardware, software, firmware, information/data, and telecommunications).”

Three key objectives that are at the heart of computer security:

1. Confidentiality: This term covers two related concepts:
 - Data confidentiality: Assures that private or confidential information is not made available or disclosed to unauthorized individuals.
 - Privacy: Assures that individuals control or influence what information related to them may be collected and stored and by whom and to whom that information may be disclosed.

2. Integrity: This term covers two related concepts:

- Data integrity: Assures that information and programs are changed only in a specified and authorized manner.
- System integrity: Assures that a system performs its intended function in an unimpaired manner, free from deliberate or inadvertent unauthorized manipulation of the system.

3. Availability: Assures that systems work promptly and service is not denied to authorized users.

❖ Called as CIA traid

Two additional concepts :

1. Authenticity: The property of being genuine and being able to be verified and trusted.
 - Confidence in the validity of a transmission, a message, or message originator.
 - Verifying that users are who they say they are and that each input arriving at the system came from a trusted source.
2. Accountability: The security goal that generates the requirement for actions of an entity to be traced uniquely to that entity.
 - Includes nonrepudiation, deterrence, fault isolation, intrusion detection and prevention, and afteraction recovery and legal action.

The OSI security architecture

Provides abstract overview on security attacks, mechanisms, and services

- **Security attack:** Any action that compromises the security of information owned by an organization.
- **Security mechanism:** A process (or a device incorporating such a process) that is designed to detect, prevent, or recover from a security attack.
- **Security service:** A processing or communication service that enhances the security of the data processing systems and the information transfers of an organization.

Attack and Threat

- **Threat** :A potential for violation of security, which exists when there is a circumstance, capability, action, or event that could breach security and cause harm.
- A threat is a possible danger that might exploit a vulnerability.
- **Attack**: The attack is an intended unauthorized action on a system/asset. An attack always has a motivation to misuse system and generally wait for an opportunity to occur.

Types of Attacks: Active and Passive

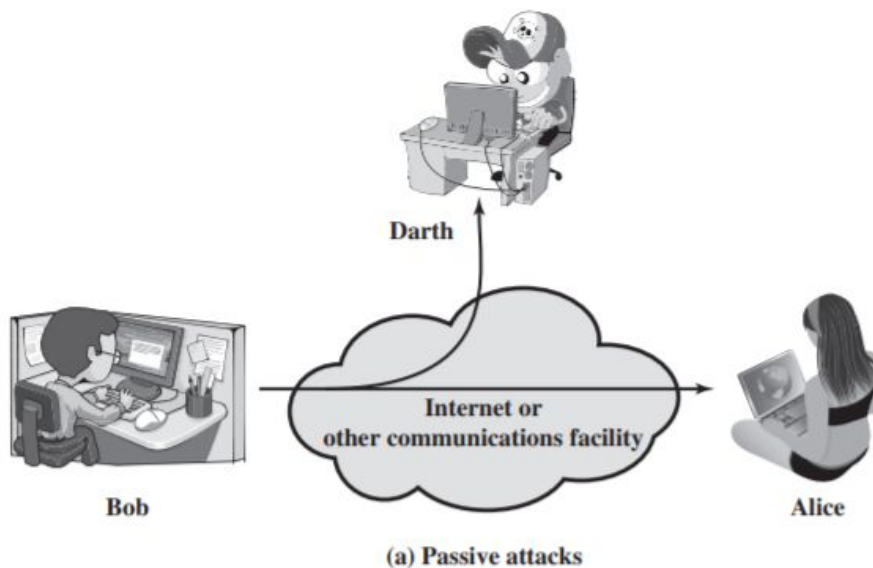
- A passive attack attempts to learn or make use of information from the system but does not affect system resources.
- An active attack attempts to alter system resources or affect their operation.

Passive Attacks :

- The nature of eavesdropping on, or monitoring of, transmissions.
- The goal of the opponent is to obtain information that is being transmitted.
- Two types of passive attacks are the *release of message contents* and *traffic analysis*.

Passive Attacks :

- Passive attacks are very difficult to detect, because they do not involve any alteration of the data.
- It is feasible to prevent the success of these attacks, usually by means of encryption.
- Thus, the emphasis in dealing with passive attacks is on prevention rather than detection



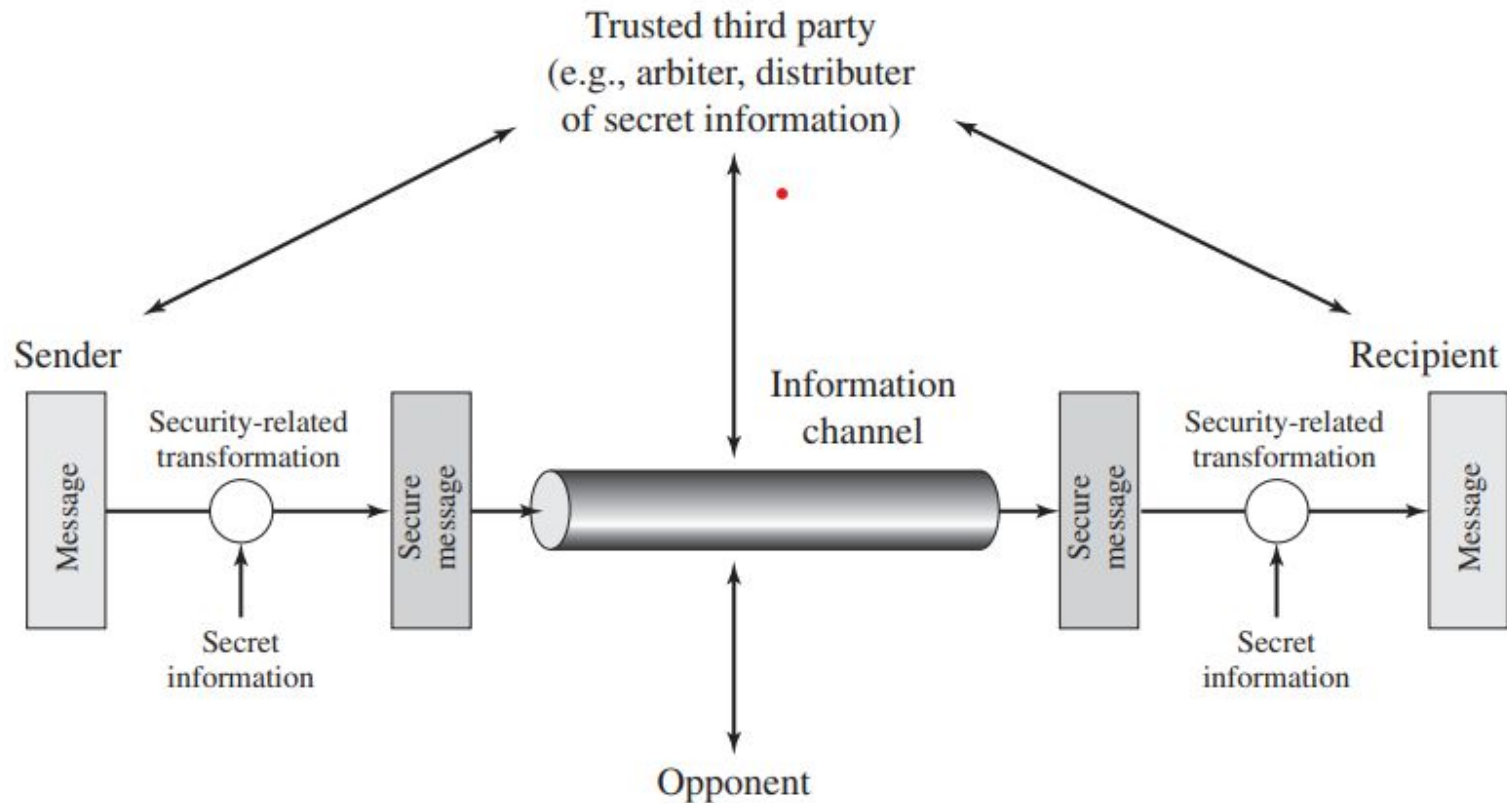
Active Attacks:

- Involve some modification of the data stream or the creation of a false stream and can be subdivided into four categories: masquerade, replay, modification of messages, and denial of service
- A masquerade takes place when one entity pretends to be a different entity.
- Replay involves the passive capture of a data unit and its subsequent retransmission to produce an unauthorized effect.
- Modification of messages simply means that some portion of a legitimate message is altered, or that messages are delayed or reordered, to produce an unauthorized effect.
- The denial of service prevents or inhibits the normal use or management of communications facilities

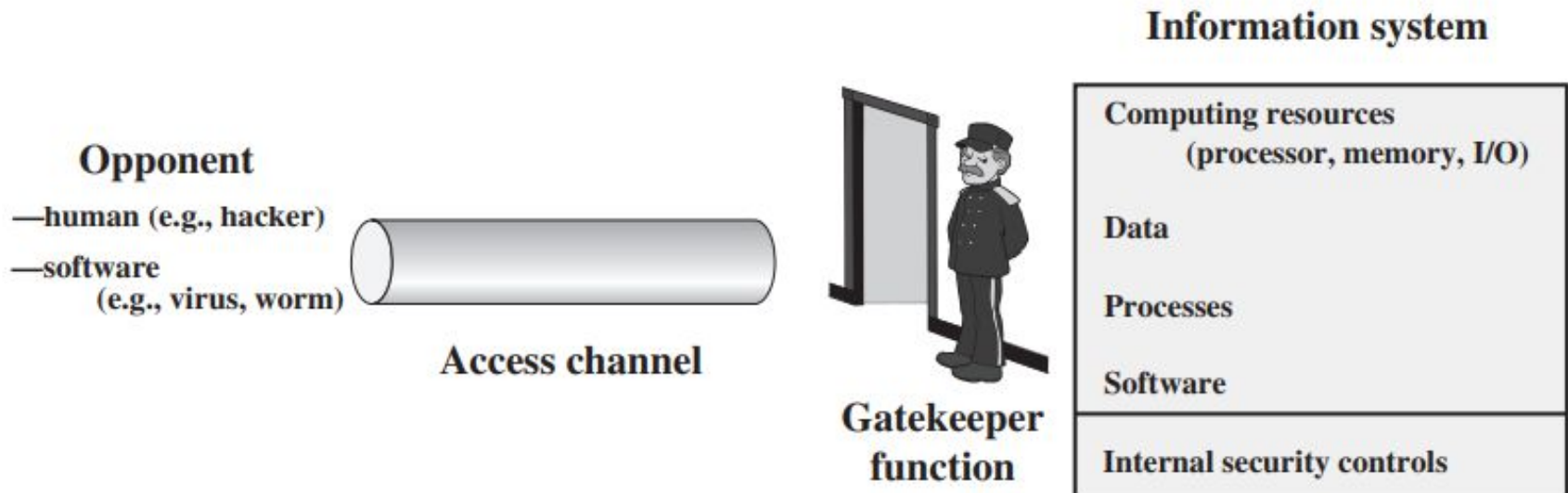
Encryption

- An original message is known as the plaintext, while the coded message is called the ciphertext.
- The process of converting from plaintext to ciphertext is known as enciphering or encryption; restoring the plaintext from the ciphertext is deciphering or decryption.
- Suite of algorithms used for encryption constitute the area of study known as cryptography.
- Such a scheme is known as a cryptographic system or a cipher.
- Techniques used for deciphering a message without any knowledge of the enciphering details fall into the area of cryptanalysis.
- Cryptanalysis is what the layperson calls “breaking the code.”
- The areas of cryptography and cryptanalysis together are called cryptology.

Model for Network Security



Network Security Access Model



- Viruses and worms are two examples of software attacks. Such attacks can be introduced into a system by means of a disk that contains the unwanted logic concealed in otherwise useful software.
- The security mechanisms needed to cope with unwanted access fall into two broad categories:
 - Gate Keeper Functions: password-based login procedures.
 - Internal controls that monitor activity and analyze stored information in an attempt to detect the presence of unwanted intruders.

Encryption

- An original message is known as the plaintext, while the coded message is called the ciphertext.
- The process of converting from plaintext to ciphertext is known as enciphering or encryption; restoring the plaintext from the ciphertext is deciphering or decryption.
- Suite of algorithms used for encryption constitute the area of study known as cryptography.
- Such a scheme is known as a cryptographic system or a cipher.
- Techniques used for deciphering a message without any knowledge of the enciphering details fall into the area of cryptanalysis.
- Cryptanalysis is what the layperson calls “breaking the code.”
- The areas of cryptography and cryptanalysis together are called cryptology.

Symmetric Cipher Model

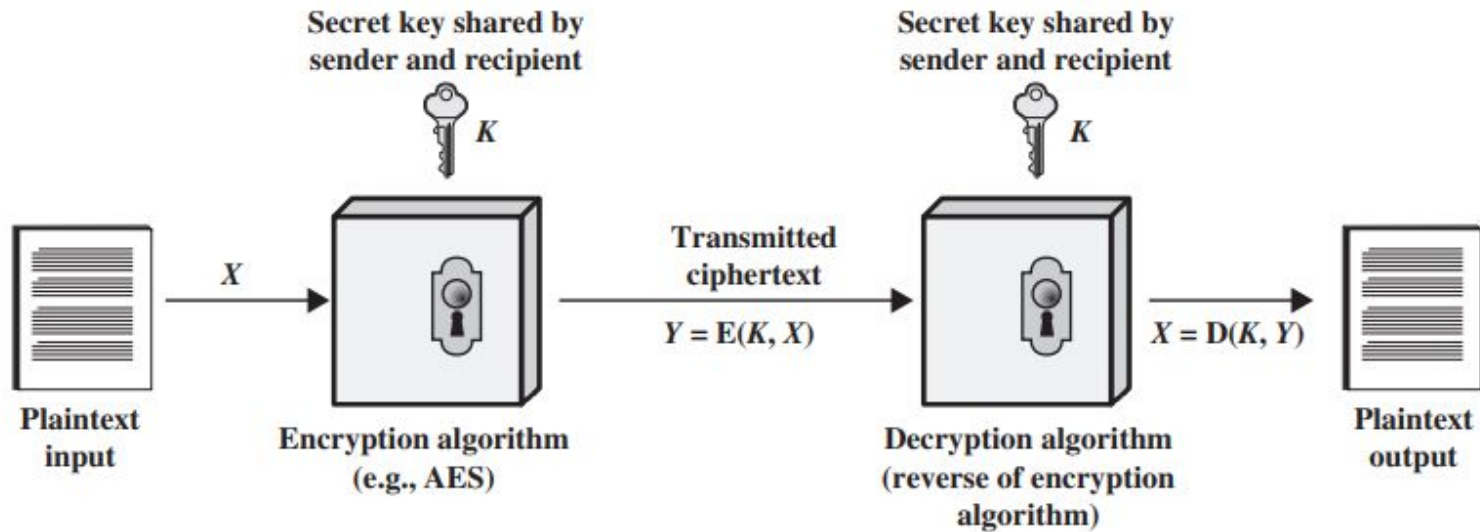


Figure 3.4 Simplified Model of Symmetric Encryption

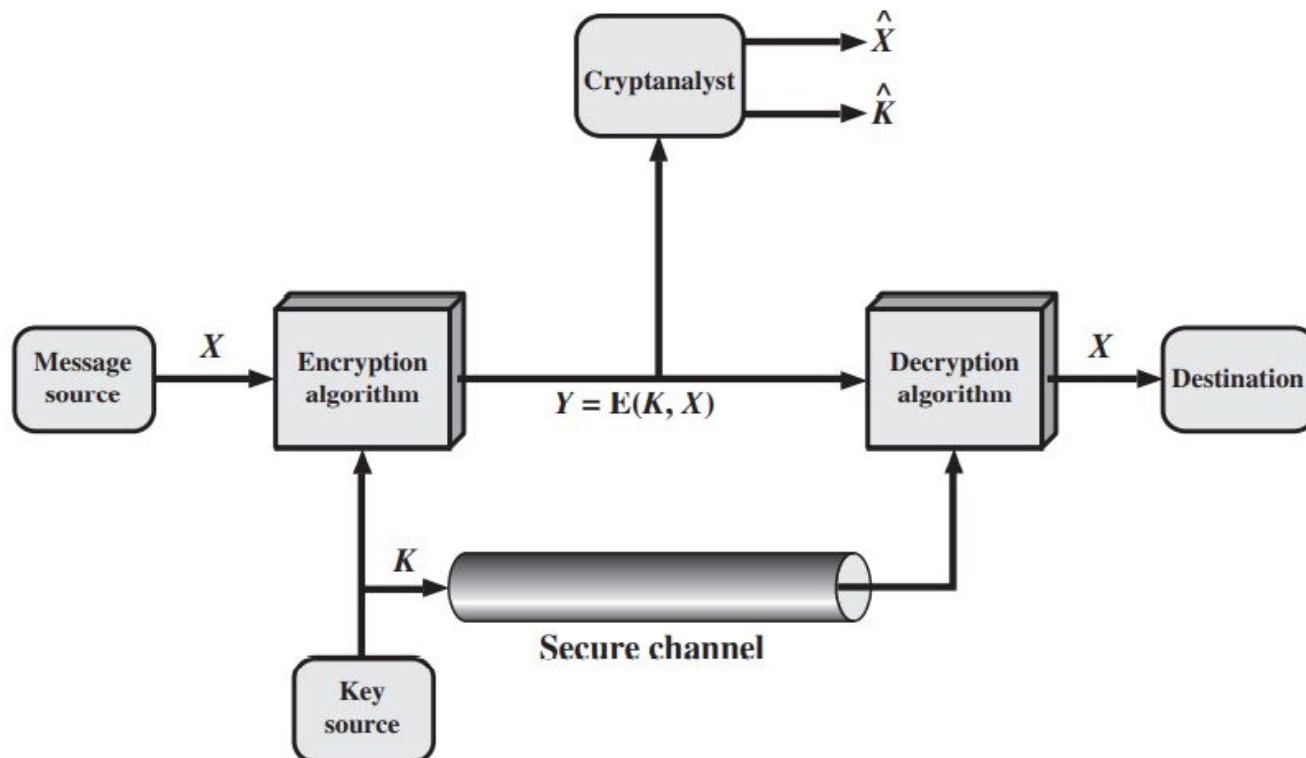
A symmetric encryption scheme has five ingredients

- Plaintext: This is the original intelligible message or data that is fed into the algorithm as input.
- Encryption algorithm: The encryption algorithm performs various substitutions and transformations on the plaintext.
- Secret key: The secret key is also input to the encryption algorithm.
- Ciphertext: This is the scrambled message produced as output.
- Decryption algorithm: This is essentially the encryption algorithm run in reverse. It takes the ciphertext and the secret key and produces the original plaintext.

Requirements of Encryption Algorithms:

1. The algorithms must be very strong such that an opponent who knows the algorithm and has access to one or more ciphertexts would be unable to decipher the ciphertext or figure out the key.
2. Sender and receiver must have obtained copies of the secret key in a secure fashion and must keep the key secure.

Modified Symmetric Model



Cryptography

Cryptographic systems are characterized along three independent dimensions:

1. **The type of operations used for transforming plaintext to ciphertext.** substitution, in which each element in the plaintext (bit, letter, group of bits or letters) is mapped into another element, and transposition, in which elements in the plaintext are rearranged.

2. **The number of keys used.** symmetric, single-key, secret-key, or conventional Encryption or Asymmetric, two-key, or public-key encryption.

3. **The way in which the plaintext is processed.** A block cipher processes the input one block of elements at a time, producing an output block for each input block. A stream cipher processes the input elements continuously, producing output one element at a time.

Cryptanalysis and Brute-Force Attack

- The objective of attacking an encryption system is to recover the key in use rather than simply to recover the plaintext of a single ciphertext.
- There are two general approaches to attacking a conventional encryption scheme:
 - Cryptanalysis: Exploits the characteristics of the algorithm to attempt to deduce a specific plaintext or to deduce the key being used.
 - Brute-force attack: Try every possible key on a piece of ciphertext until an intelligible translation into plaintext is obtained.

Table 2.1 Types of Attacks on Encrypted Messages

Type of Attack	Known to Cryptanalyst
Ciphertext Only	• Encryption algorithm • Ciphertext
Known Plaintext	• Encryption algorithm • Ciphertext • One or more plaintext–ciphertext pairs formed with the secret key
Chosen Plaintext	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen Ciphertext	• Encryption algorithm • Ciphertext • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

Building blocks of Encryption techniques

1. Substitution: A substitution technique is one in which the letters of plaintext are replaced by other letters or by numbers or symbols.
2. Transposition: permutation on the plaintext letters.

Substitution Methods:

- Caesar Cipher.
- Monoalphabetic Ciphers.
 - Playfair Cipher
 - Hill Cipher
- Polyalphabetic Ciphers
 - Vigenère Cipher
 - Vernam Cipher

Caesar Cipher.

- The earliest known, and the simplest, use of a substitution cipher.

plain: meet me after the toga party

cipher: PHHW PH DIWHU WKH WRJD SDUWB

- The algorithm can be expressed as follows:
- For each plaintext letter P, substitute the ciphertext letter C:

$$C = E(3, p) = (p + 3) \bmod 26$$

In general,

$C = E(k, p) = (p + k) \bmod 26$Encryption.

$p = D(k, C) = (C - k) \bmod 26$Decryption.

****Very much prone to Brute force attack.**

Monoalphabetic Ciphers.

- With only 25 possible keys, the Caesar cipher is far from secure.
 - Increase in the key space can be achieved by allowing an arbitrary substitution.
 - A permutation of a finite set of elements S is an ordered sequence of all the elements of S , with each element appearing exactly once.
 - For example, if $S = \{a, b, c\}$, there are six permutations of S : abc , acb , bac , bca , cab , cba .
 - In general, there are $n!$ permutations of a set of n elements.(26! Possible substitutions)
- ** Brute force attack is difficult, but prone to frequency analysis attack.**

Monoalphabetic Ciphers.

- With only 25 possible keys, the Caesar cipher is far from secure.
- Increase in the key space can be achieved by allowing an arbitrary substitution.
- A permutation of a finite set of elements S is an ordered sequence of all the elements of S , with each element appearing exactly once.
- For example, if $S = \{a, b, c\}$, there are six permutations of S : $abc, acb, bac, bca, cab, cba$.
- In general, there are $n!$ permutations of a set of n elements.(26! Possible substitutions)

**** Brute force attack is difficult, but prone to frequency analysis attack- Diagrams/**

Homophones

Playfair cipher

- Treats digrams in the plaintext as single units and translates these units into ciphertext digrams.
- The Playfair algorithm is based on the use of a 5 * 5 matrix of letters constructed using a keyword.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Plaintext is encrypted two letters at a time, according to the following rules:

1. Repeating plaintext letters that are in the same pair are separated with a filler letter, such as x, so that balloon would be treated as ba lx lo on.
2. Two plaintext letters that fall in the same row of the matrix are each replaced by the letter to the right, with the first element of the row circularly following the last. For example, ar is encrypted as RM.
3. Two plaintext letters that fall in the same column are each replaced by the letter beneath, with the top element of the column circularly following the last. For example, mu is encrypted as CM.
4. Otherwise, each plaintext letter in a pair is replaced by the letter that lies in its own row and the column occupied by the other plaintext letter. Thus, hs becomes BP and ea becomes IM (or JM, as the encipherer wishes).

Transposition Techniques

- A kind of mapping is achieved by performing some sort of permutation on the plaintext letters.
- This technique is referred to as a transposition cipher.

Rail fence

For example, to encipher the message “meet me after the toga party” with a rail fence of depth 2:

m e m a t r h t g p r y
e t e f e t e o a a t

The encrypted message is

MEMATRHTGPRYETEFETEOAAT

Transposition Techniques

- A more complex scheme is to write the message in a rectangle, row by row, and read the message off, column by column, but permute the order of the columns.
- The order of the columns then becomes the key to the algorithm.

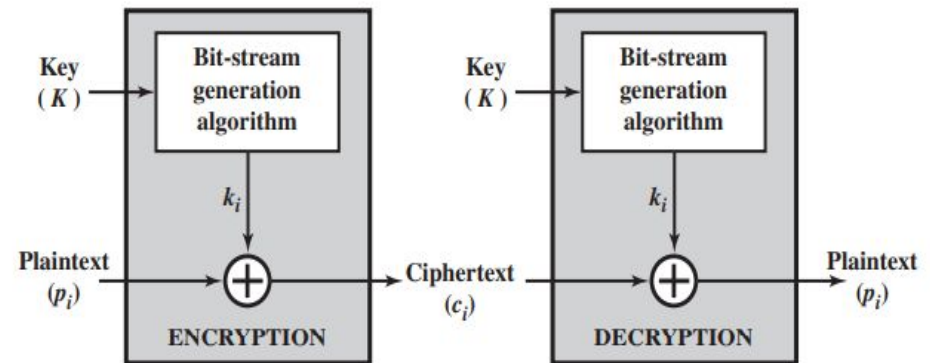
Key:	4	3	1	2	5	6	7
Plaintext:	a	t	t	a	c	k	p
	o	s	t	p	o	n	e
	d	u	n	t	i	l	t
	w	o	a	m	x	y	z
Ciphertext:	T	T	N	A	P	T	M
	T	S	U	O	A	O	D
	W	C	O	I	X	K	N
	L	Y	P	E	T	Z	

- The key is 4312567.
- Encryption-- start with the column that is labeled 1, in this case column 3. Write down all the letters in that column. Proceed to column 4, which is labeled 2, then column 2, then column 1, then columns 5, 6, and 7

Traditional Block Cipher Structure

Stream ciphers and Block ciphers

- A stream cipher is one that encrypts a digital data stream one bit or one byte at a time.
- The keystream (k_i) is as long as the plaintext bit stream (p_i).
- If the cryptographic keystream is random, then this cipher is unbreakable by any means other than acquiring the keystream.
- The keystream must be provided to both users in advance via some independent and secure channel.
- This leads to logistical problems if the intended data traffic is very large

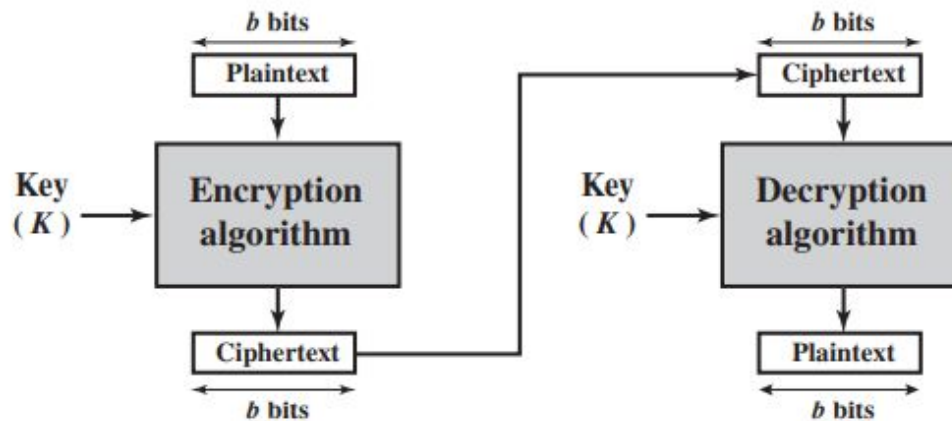


(a) Stream cipher using algorithmic bit-stream generator

Traditional Block Cipher Structure

Stream ciphers and Block ciphers

- A block cipher is one in which a block of plaintext is treated as a whole and used to produce a ciphertext block of equal length.
- Typically, a block size of 64 or 128 bits is used.
- Block ciphers have broader range of applications than stream ciphers.



(b) Block cipher

Feistel Cipher Structure

Motivation for the Feistel Cipher Structure.

- A block cipher operates on a plaintext block of n bits to produce a ciphertext block of n bits.
- There are 2^n possible different plaintext blocks.
- For the encryption to be reversible each must produce a unique ciphertext block—
Reversible(Non Singular) transformation.
- For $n=2$,

Reversible Mapping	
Plaintext	Ciphertext
00	11
01	10
10	00
11	01

Irreversible Mapping	
Plaintext	Ciphertext
00	11
01	10
10	01
11	01

Block ciphers

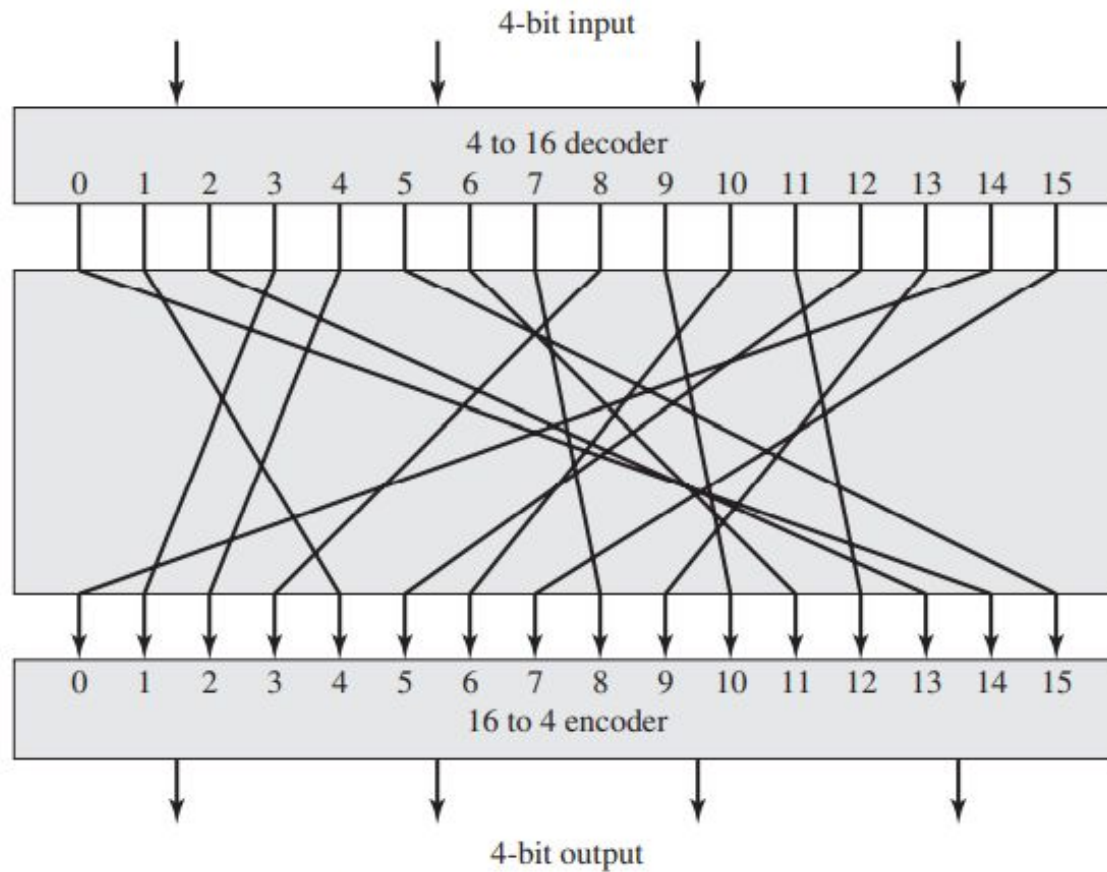


Figure 3.2 General n -bit- n -bit Block Substitution (shown with $n = 4$)

Block ciphers

Table 3.1 Encryption and Decryption Tables for Substitution Cipher of Figure 3.2

Plaintext	Ciphertext
0000	1110
0001	0100
0010	1101
0011	0001
0100	0010
0101	1111
0110	1011
0111	1000
1000	0011
1001	1010
1010	0110
1011	1100
1100	0101
1101	1001
1110	0000
1111	0111

Ciphertext	Plaintext
0000	1110
0001	0011
0010	0100
0011	1000
0100	0001
0101	1100
0110	1010
0111	1111
1000	0111
1001	1101
1010	1001
1011	0110
1100	1011
1101	0010
1110	0000
1111	0101

Block ciphers

- Feistel refers to this as the ideal block cipher, because it allows for the maximum number of possible encryption mappings from the plaintext block.
- If the block size(n) is small--- Brute force attack(Ceaser cipher).
- With n being large, and encryption is reversible, the statistical characteristics of the source plaintext are masked to such an extent that this type of cryptanalysis is infeasible.
- For transformations to be performed, the mapping is defined by the key.
- It is the key that determines the specific mapping from among all possible mappings.

Feistel Cipher Structure

- In this method, the block cipher is obtained by the concept of a product cipher.
- It means execution of two or more simple ciphers in sequence in such a way that the final result or product is cryptographically stronger than any of the component ciphers.
- The purpose of the approach is to develop a block cipher with a key length of k bits and a block length of n bits, allowing a total of 2^k possible transformations.

Feistel Cipher

- Feistel cipher algorithm alternates substitutions and permutations.
 - Substitution: Each plaintext element or group of elements is uniquely replaced by a corresponding ciphertext element or group of elements.
 - Permutation: A sequence of plaintext elements is replaced by a permutation of that sequence. That is, no elements are added or deleted or replaced in the sequence, rather the order in which the elements appear in the sequence is changed.

Product Cipher

- Alternates between confusion and diffusion functions.
- The intension is to prevent cryptanalysis based on statistical analysis.
- The attacker has some knowledge of the statistical characteristics of the plaintext.
 - In a human-readable message in some language.
 - The frequency distribution of the various letters may be known.
 - Words or phrases likely to appear in the message (probable words).
- If these statistics appear in ciphertext, cryptanalyst- deduce key or plaintext

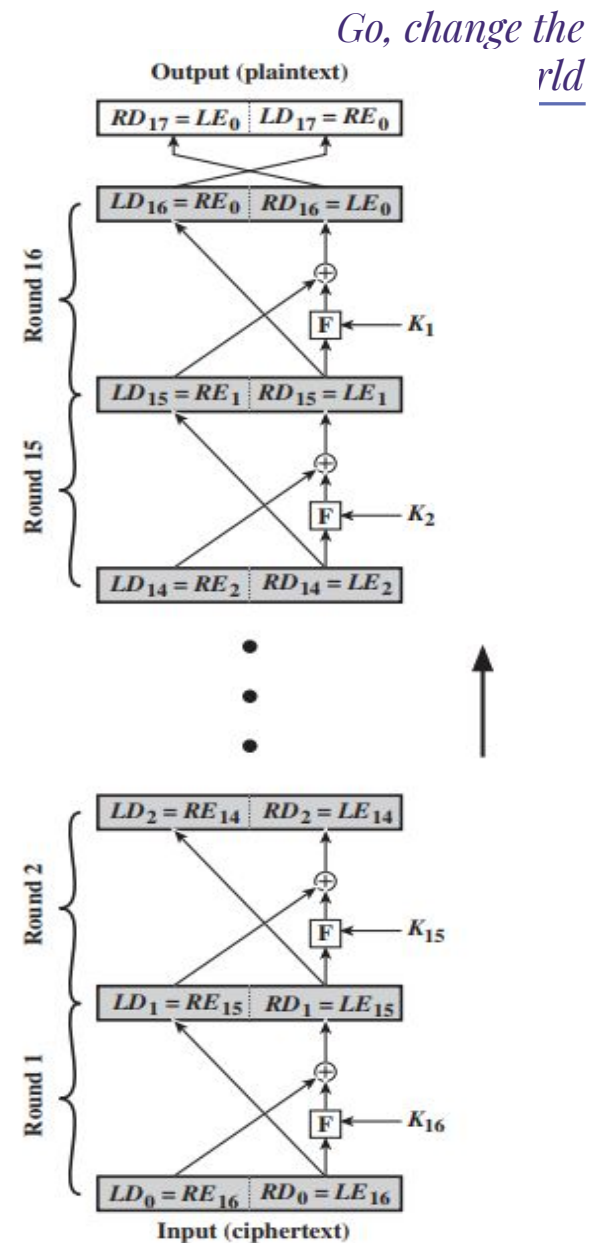
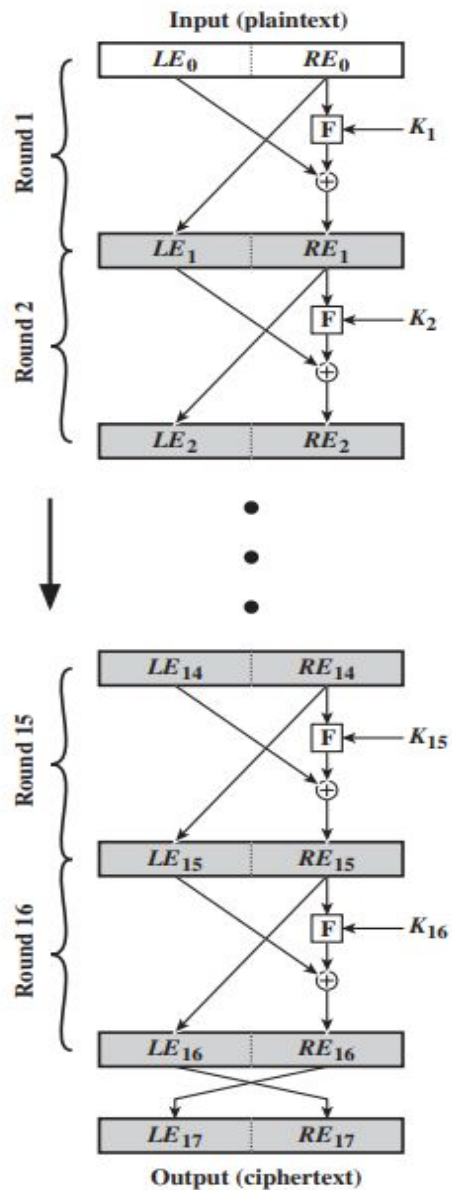
Confusion and Diffusion function

- Methods for frustrating statistical cryptanalysis.
- In diffusion, the statistical structure of the plaintext is dissipated into long-range statistics of the ciphertext.
- Achieved by having each plaintext digit affect the value of many ciphertext digits.
- Each ciphertext digit be affected by many plaintext digits.
- In a binary block cipher, diffusion can be achieved by repeatedly performing some permutation on the data followed by applying a function to that permutation.
- The effect is that bits from different positions in the original plaintext contribute to a single bit of ciphertext

Confusion and Diffusion function

- Every block cipher involves a transformation of a block of plaintext into a block of ciphertext, where the transformation depends on the key.
- **Diffusion** seeks to make the statistical relationship between the plaintext and ciphertext as complex as possible in order to thwart attempts to deduce the key.
- **Confusion** seeks to make the relationship between the statistics of the ciphertext and the value of the encryption key as complex as possible, again to thwart attempts to discover the key.
- If the attacker can get some handle on the statistics of the ciphertext, the way in which the key was used to produce that ciphertext is so complex as to make it difficult to deduce the key.

Structure of Feistel Cipher.



*Go, change the
rld*

- The inputs to the encryption algorithm are a plaintext block of length $2w$ bits and a key K .
- The plaintext block is divided into two halves, L_0 and R_0 .
- The two halves of the data pass through n rounds of processing and then combine to produce the ciphertext block.
- Each round i has as inputs L_{i-1} and R_{i-1} derived from the previous round, as well as a subkey K_i derived from the overall K .
- In general, the subkeys K_i are different from K and from each other.
- There are 16 rounds and all the rounds have same structure.
- A substitution is performed on the left half of the data- By applying round function F to right half of the data and taking XOR of output of that function and left half of the data.

- F is a function of right-half block of w bits and a subkey of y bits, which produces an output value of length w bits:

$$F(RE_i, K_{i+1}).$$

- Following this substitution, a permutation is performed that consists of the interchange of the two halves of the data.
- Feistel is therefore substitution- permutation network (SPN).

- The realization of the Feistel cipher depends on the following parameters:
 1. Block size: Larger block sizes mean greater security but reduced encryption/decryption speed for a given algorithm. The greater security is achieved by greater diffusion. (Usually block size is 64bits).
 2. Key size: Larger key size means greater security but may decrease encryption/decryption speed. The greater security is achieved by greater resistance to brute-force attacks and greater confusion.(Usually 128 bits).
 3. Number of rounds: The essence of the Feistel cipher is that a single round offers inadequate security but that multiple rounds offer increasing security.A typical size is 16 rounds.

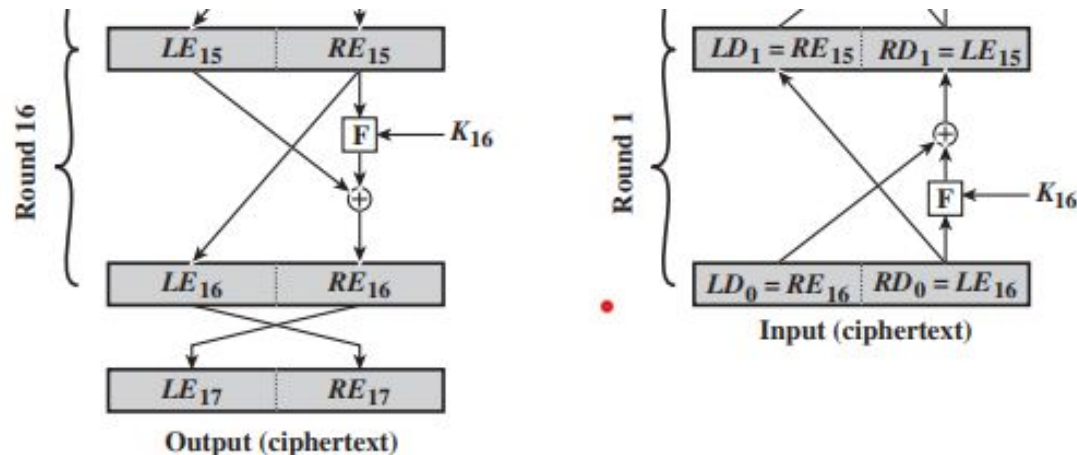
4. Subkey generation algorithm: Greater complexity in this algorithm should lead to greater difficulty of cryptanalysis.
5. Round function F: Again, greater complexity generally means greater resistance to cryptanalysis.
6. Fast software encryption/decryption: Encryption is embedded in applications or utility functions in such a way as to preclude a hardware implementation.
7. Ease of analysis: if the algorithm can be concisely and clearly explained, it is easier to analyze that algorithm for cryptanalytic vulnerabilities and therefore develop a higher level of assurance as to its strength

Feistel Decryption Algorithm

- The ciphertext is used as input to the algorithm and the subkeys K_i are used in reverse order.
- K_n in the first round, K_{n-1} in the second round, and so on, until K_1 is used in the last round.
- If LE_i and RE_i are used for data traveling through the encryption algorithm then LD_i and RD_i are used for data traveling through the decryption algorithm.
- Let the output of the i th encryption round be $LE_i \parallel RE_i$ (LE_i concatenated with RE_i). Then the corresponding output of the $(16 - i)$ th decryption round is $RE_i \parallel LE_i$ or, equivalently, $LD_{16-i} \parallel RD_{16-i}$

Feistel cipher realization

- After the last iteration of the encryption process, the two halves of the output are swapped, so that the ciphertext is $RE_{16} \parallel LE_{16}$.
- Now take that ciphertext and use it as input to the same algorithm for decryption.
- The input to the first round is $RE_{16} \parallel LE_{16}$, which is equal to the 32-bit swap of the output of the sixteenth round of the encryption process



Feistel cipher realization

Consider the encryption process.

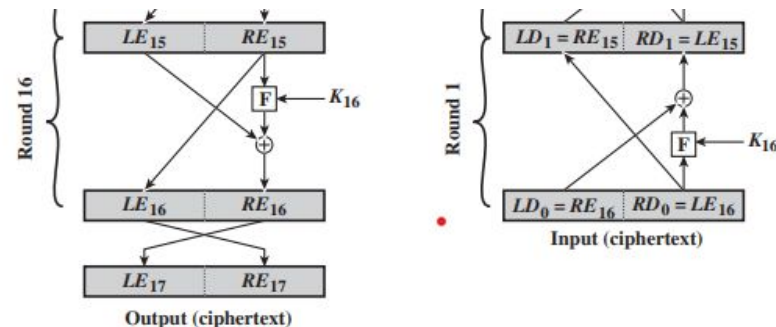
- $LE_{16} = RE_{15}$
- $RE_{16} = LE_{15} \oplus F(RE_{15}, K_{16})$
- $LD_1 = RD_0 = LE_{16} = RE_{15}$
- $RD_1 = LD_0 \oplus F(RD_0, K_{16})$

$$= RE_{16} \oplus F(RE_{15}, K_{16})$$

$$= [LE_{15} \oplus F(RE_{15}, K_{16})] \oplus F(RE_{15}, K_{16}).$$

$$= LE_{15}.$$

$$\underline{LD_1 = RE_{15} \text{ and } RD_1 = LE_{15}}$$



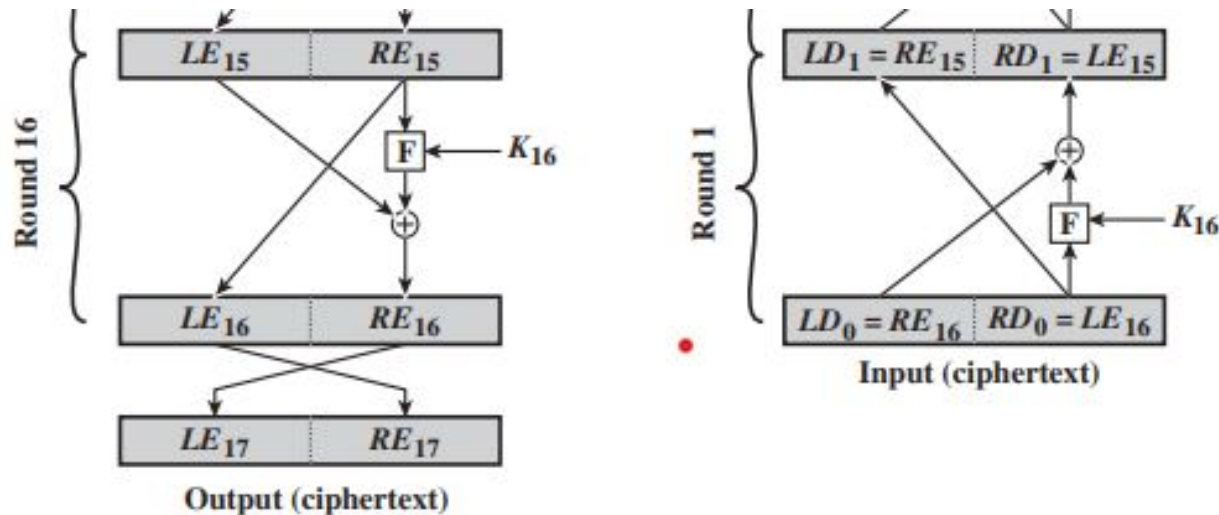
The XOR has the following properties:

$$[A \oplus B] \oplus C = A \oplus [B \oplus C]$$

$$D \oplus D = 0$$

$$E \oplus 0 = E$$

- The output of the first round of the decryption process is $RE_{15} || LE_{15}$, which is the 32-bit swap of the input to the sixteenth round of the encryption.
- This correspondence holds all the way through the 16 iterations



In General,

- $LE_i = RE_{i-1}$
- $RE_i = LE_{i-1} \oplus F(RE_{i-1}, K_i)$

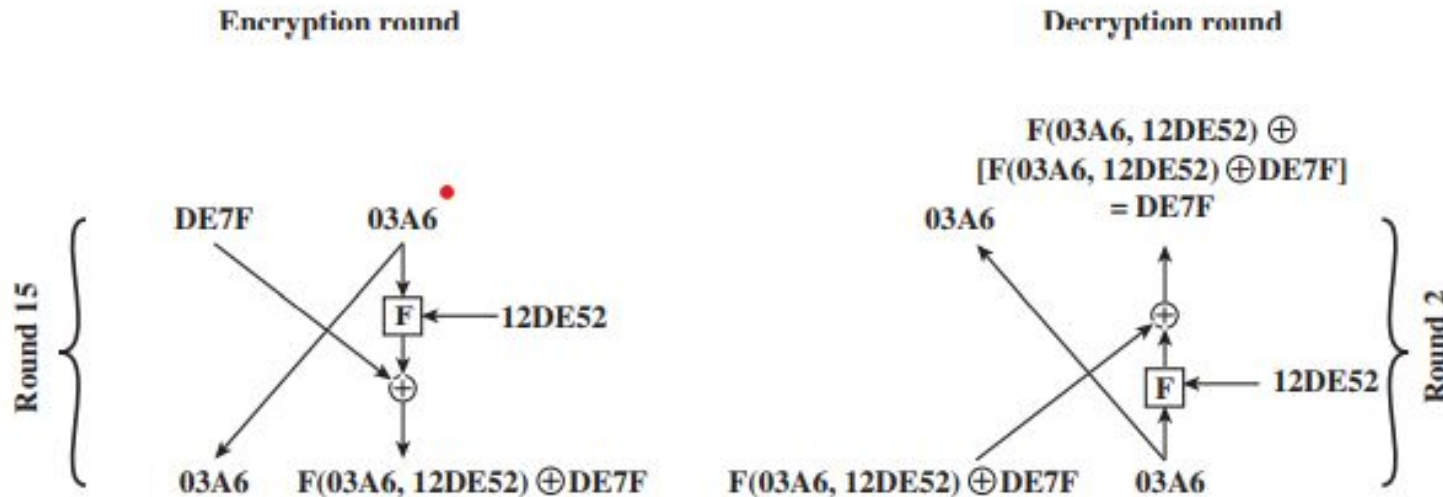
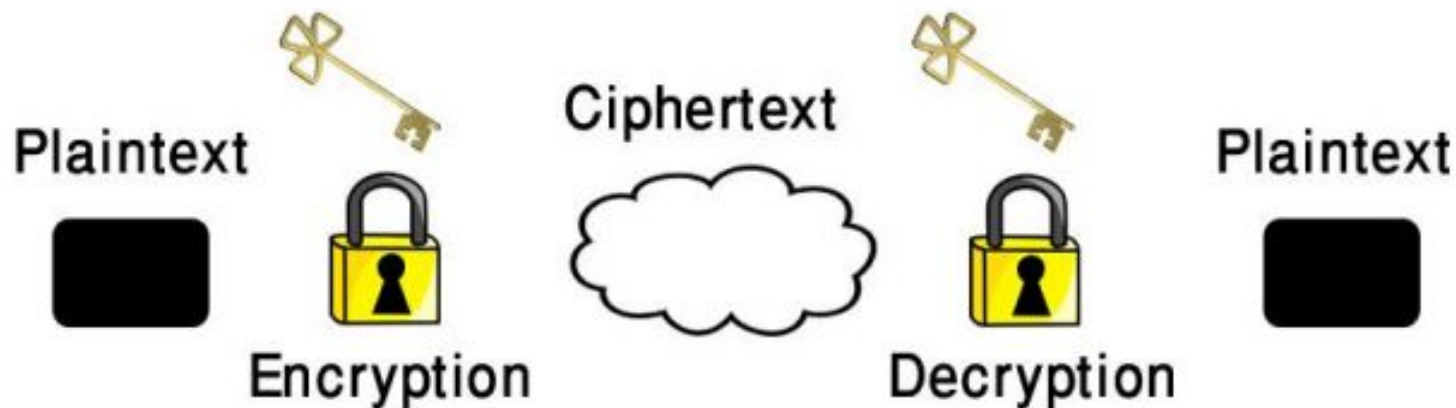


Figure 3.4 Feistel Example

- $LE14 = DE7F$ and $RE14 = 03A6$.
- $K15$ is $12DE52$
- After round 15, we have $LE15 = 03A6$ and $RE15 = F(03A6, 12DE52) \oplus DE7F$.
- $LD1 = RE15$ and $RD1 = LE15$, (Need that $LD2 = RE14$ and $RD2 = LE14$).
- $LD1 = F(03A6, 12DE52) \oplus DE7F$ and $RD1 = 03A6$.
- $LD2 = 03A6 = RE14$.
- $RD2 = F(03A6, 12DE52) \oplus [F(03A6, 12DE52) \oplus DE7F] = DE7F = LE14$.

Stream Ciphers and Block Ciphers, Feistel Cipher Structure.

- Encryption Definition: –The action of disguising information so that it can be recovered easily by the persons who have the key, but is highly resistant to recovery by persons who do not have the key.



- A message is cleartext (plaintext) is encrypted (disguised) through the use of an encryption key to create a Ciphertext.
- The encryption key may be changed from time to time to make an intruder's task more difficult.
- Restoration of a ciphertext to cleartext is achieved by the action of decryption using a decryption key.
- In symmetric (Single key) - The encryption and decryption keys are the same.
- In asymmetric (two keys) - The encryption and decryption keys are different.

Encryption Methods

- **Encryption Methods** - Encryption is accomplished by scrambling the bits, characters, words, or phrases in the original message.
- Scrambling involves two activities:
- **Transposition** - In which the order of the bits patterns, characters, words or phrases is rearranged.
- **Substitution** - In which new bit patterns, characters, words, or phrases are substituted for the originals without changing their order.

STREAM CIPHERS and BLOCK CIPHERS

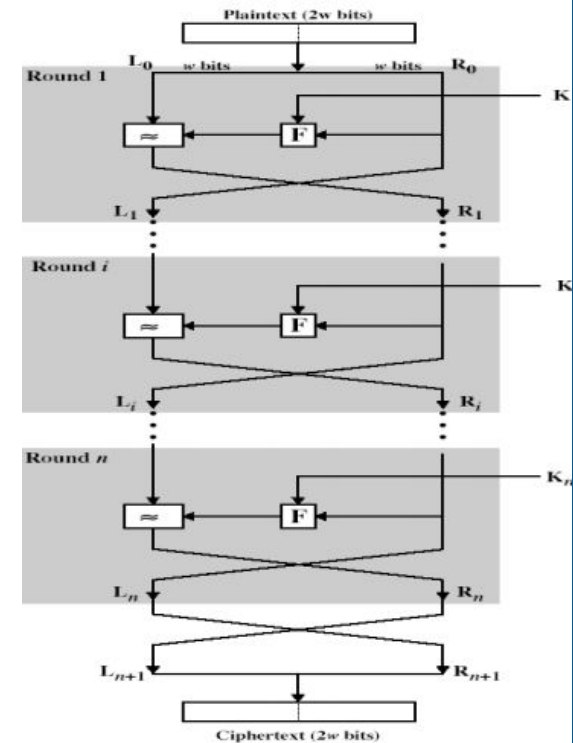
- Stream ciphers process messages a bit or byte at a time when en/decrypting.
- Block ciphers process messages into blocks, each of which is then en/decrypted – 64-bits or more
- **Many current ciphers are block ciphers** hence are focus of course

Block Cipher Principles

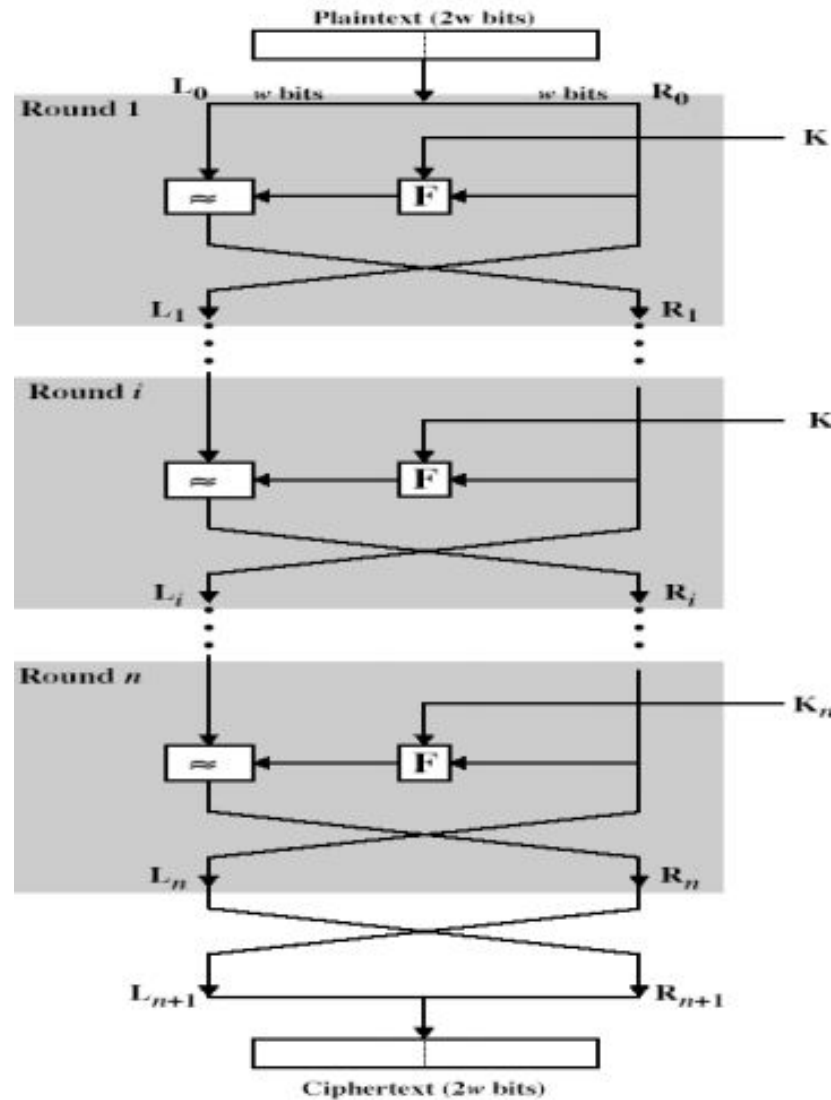
- block ciphers look like an extremely large substitution would need table of 2^{64} entries for a 64-bit block, arbitrary reversible substitution cipher for a large block size is not practical
- – 64-bit general substitution block cipher, key size 2^{64} !
- Most symmetric block ciphers are based on a **Feistel Cipher Structure**

FEISTEL CIPHER STRUCTURE

- Horst Feistel devised the feistel cipher – implements Shannon's substitution-permutation network
- Concept - partitions input block into two halves
 - process through multiple rounds which
 - perform a substitution on left data half
 - based on round function of right half & subkey
 - then have permutation swapping halves



FEISTEL CIPHER STRUCTURE



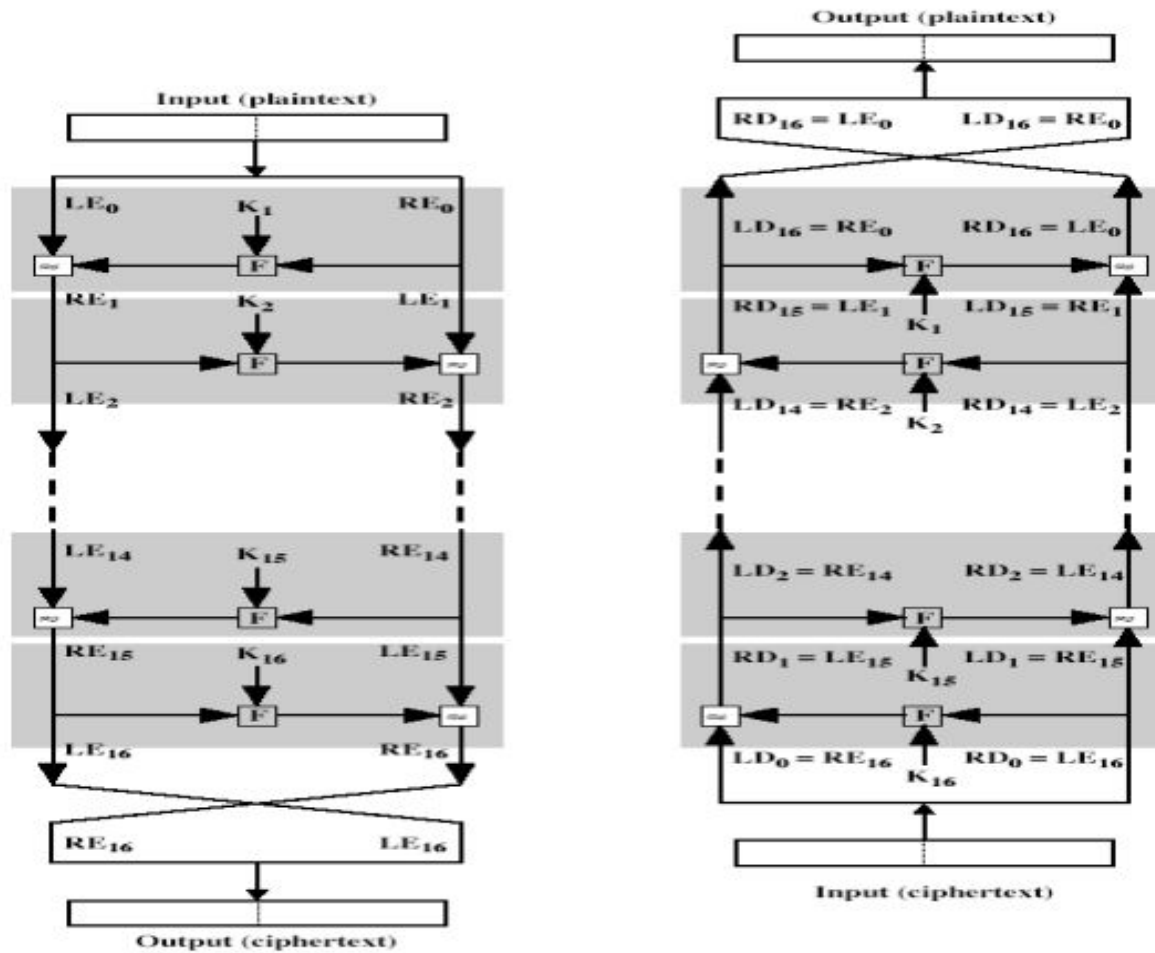
FEISTEL CIPHER

1. n sequential rounds
2. A substitution on the left half L_{i-1} .
3. Apply a round function F to the right half R_i .
4. Take XOR of the output of (1) and L_i
5. The round function is parameterized by the subkey K_i – K_i are derived from the overall key K

FEISTEL CIPHER DESIGN PRINCIPLES

- **block size** - increasing size improves security, but slows cipher
- **key size** - increasing size improves security, makes exhaustive key searching harder, but may slow cipher
- **number of rounds** - increasing number improves security, but slows cipher
- **subkey generation** - greater complexity can make analysis harder, but slows cipher
- **round function** - greater complexity can make analysis harder, but slows cipher
- **fast software en/decryption & ease of analysis** - are more recent concerns for practical use and testing

FEISTEL CIPHER DECRYPTION



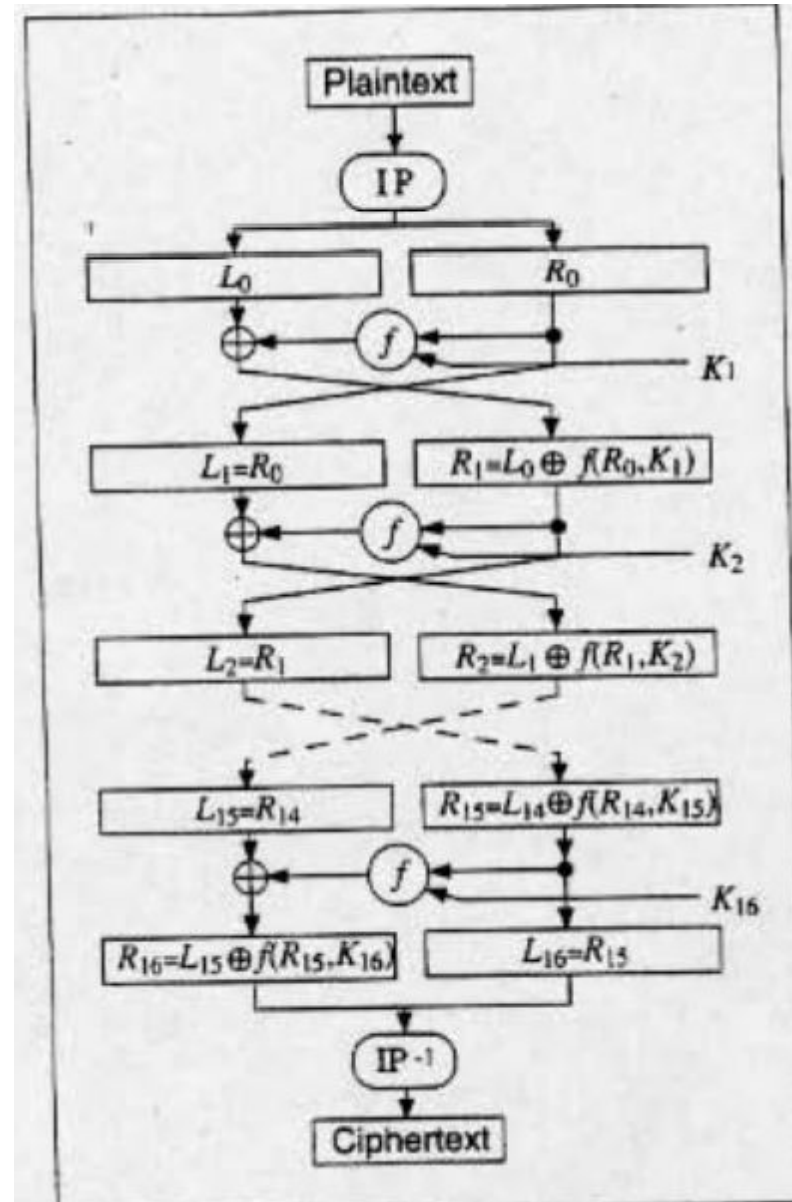
DATA ENCRYPTION STANDARD (DES)

- Most widely used block cipher in world
- Adopted in 1977 by NBS (now NIST) - as FIPS PUB 46
- Encrypts 64-bit data using 56-bit key.
- The algorithm itself is referred to as the Data Encryption Algorithm (DEA).
- The algorithm transforms 64-bit input in a series of steps into a 64-bit output.
- The same steps, with the same key, are used to reverse the encryption.

DES Algorithm

- Operates on 64 bit block of plain text.
- After Initial permutation the block is broken into right and left halves each of 32 bits long.
- Then there are 16 rounds of identical operations called Round Function- combines the data and the key.
- After 16th round the left and right round are joined and inverse permutation finishes off the algorithm.

DES Algorithm



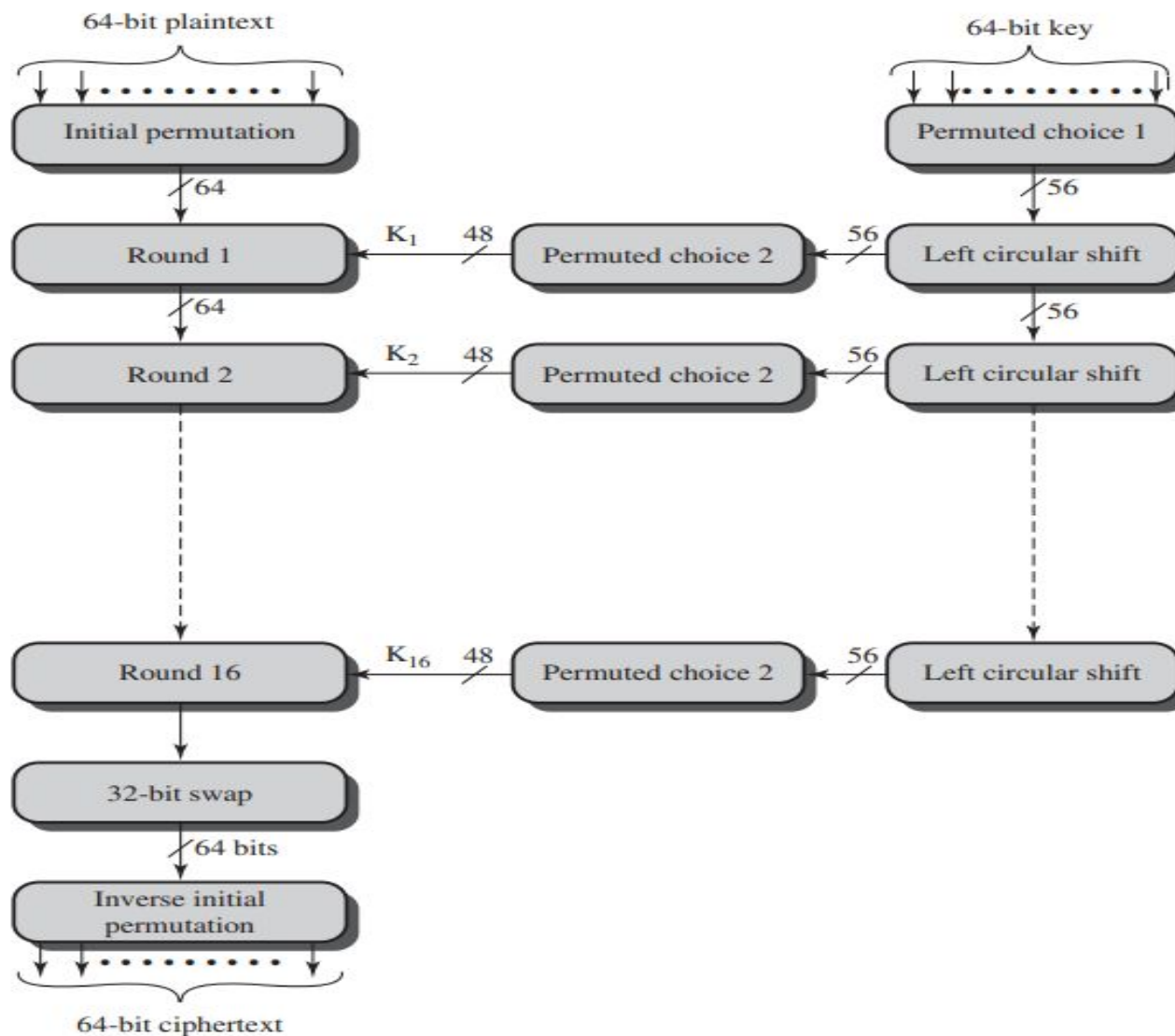
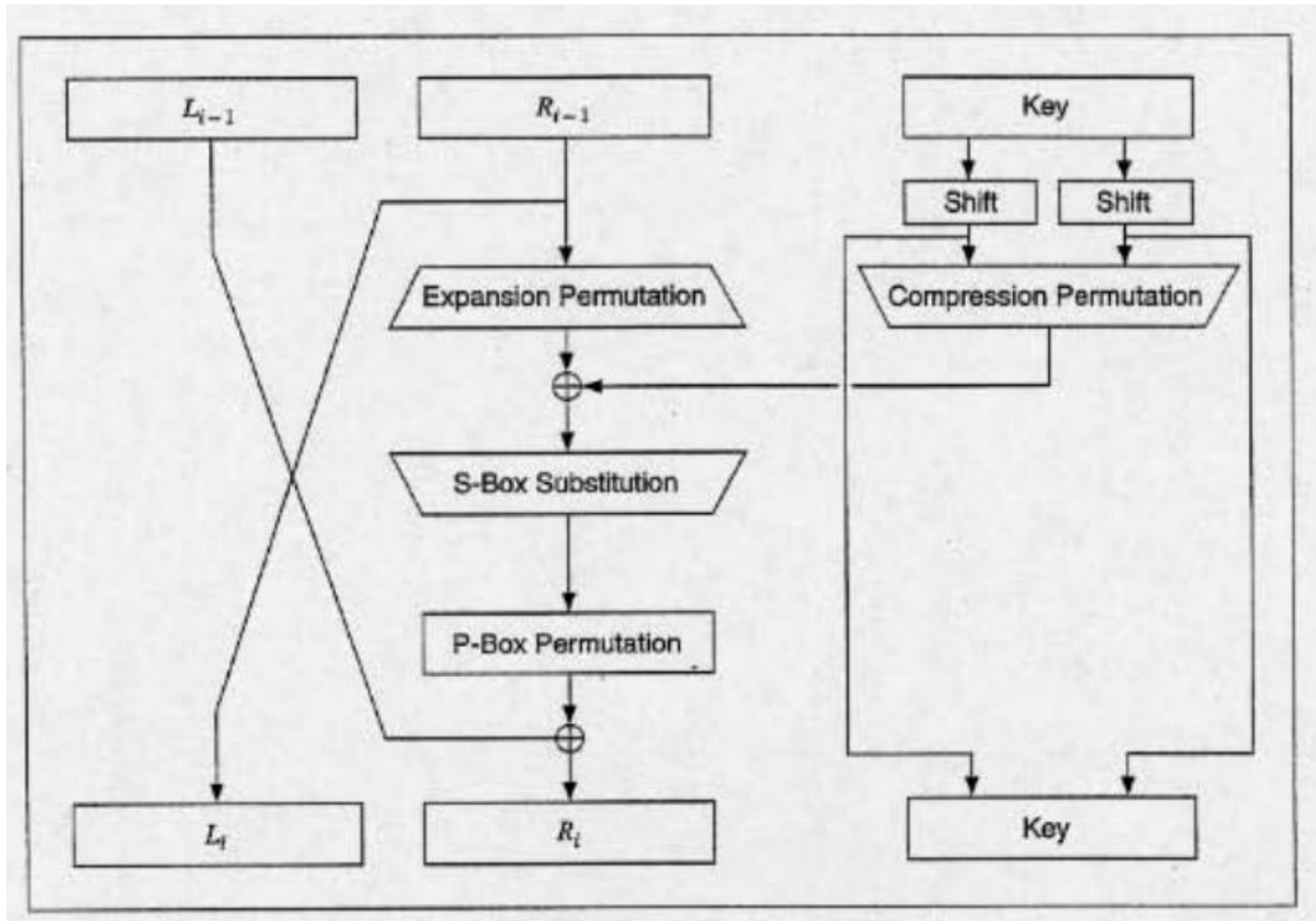


Figure 3.5 General Depiction of DES Encryption Algorithm

One round function



- There are two inputs to the encryption function: the plaintext to be encrypted and the key.
- The plaintext must be 64 bits in length and the key is 56 bits in length.
- There are three phases in encryption process:
 1. Initial Permutation: rearranges the 64-bits to produce the permuted input.
 2. Sixteen rounds- Consisting of permutation and substitution.
 3. Inverse Permutation-inverse of the initial permutation function, to produce the 64-bit ciphertext.

Key usage

- The key(56 bits) is passed through a permutation function.
- For each of the sixteen rounds, a subkey (K_i) is produced by the combination of a left circular shift and a permutation.
- The permutation function is the same for each round, but a different subkey is produced because of the repeated shifts of the key bits.

DES Decryption

- As with any Feistel cipher, decryption uses the same algorithm as encryption, except that the application of the subkeys is reversed.
- Additionally, the initial and final permutations are reversed.

Avalanche Effect

- A desirable property of any encryption algorithm is that a small change in either the plaintext or the key should produce a significant change in the ciphertext.
- Avalanche Effect: a change in one bit of the plaintext or one bit of the key should produce a change in many bits of the ciphertext.

Strength of DES

- Depends on two factors: : key size and the nature of the algorithm.
- Key size: With a key length of 56 bits, there are 256 possible keys, which is approximately 7.2×10^{16} keys. Hence Brute force attack is impossible.

Table 3.5 Average Time Required for Exhaustive Key Search

Key Size (bits)	Cipher	Number of Alternative Keys	Time Required at 10^9 Decryptions/s	Time Required at 10^{13} Decryptions/s
56	DES	$2^{56} \approx 7.2 \times 10^{16}$	2^{55} ns = 1.125 years	1 hour
128	AES	$2^{128} \approx 3.4 \times 10^{38}$	2^{127} ns = 5.3×10^{21} years	5.3×10^{17} years
168	Triple DES	$2^{168} \approx 3.7 \times 10^{50}$	2^{167} ns = 5.8×10^{33} years	5.8×10^{29} years
192	AES	$2^{192} \approx 6.3 \times 10^{57}$	2^{191} ns = 9.8×10^{40} years	9.8×10^{36} years
256	AES	$2^{256} \approx 1.2 \times 10^{77}$	2^{255} ns = 1.8×10^{60} years	1.8×10^{56} years
26 characters (permutation)	Monoalphabetic	$26! \approx 4 \times 10^{26}$	2×10^{26} ns = 6.3×10^9 years	6.3×10^6 years

Nature of DES Algorithm

- There is possibility that cryptanalysis is possible by exploiting the characteristics of the DES algorithm.
- The focus of concern has been on the eight substitution tables, or S-boxes, that are used in each iteration .
- S- boxes were constructed in such a way that cryptanalysis is possible for an opponent who knows the weaknesses in the S-boxes.

Timing Attacks- For Symmetric ciphers

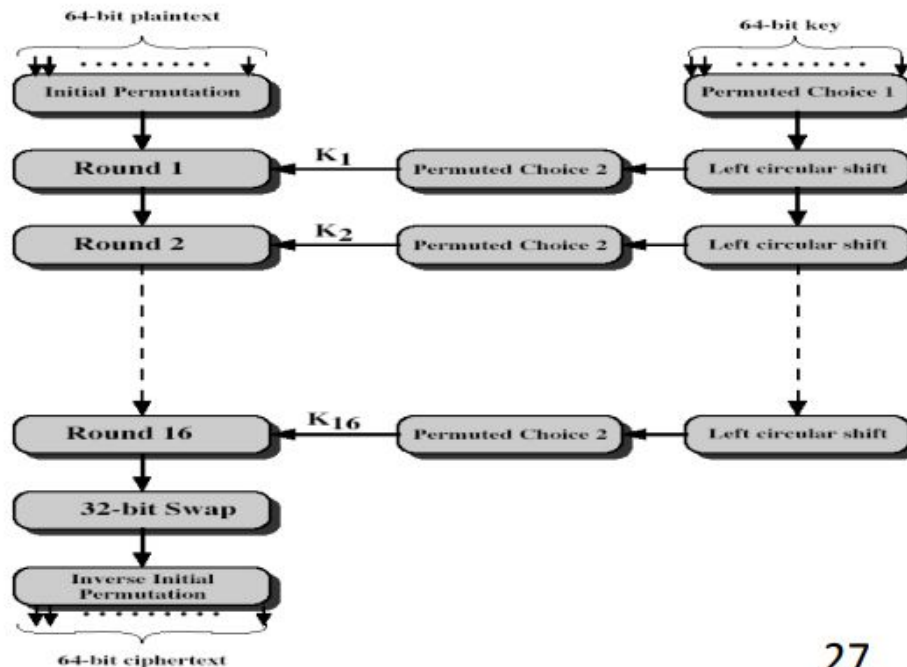
- Information about the key or the plaintext is obtained by observing how long it takes a given implementation to perform decryptions on various ciphertexts.
- A timing attack exploits the fact that an encryption or decryption algorithm often takes slightly different amounts of time on different inputs.

DES vs Triple DES and AES algorithm.

- The authors conclude that DES appears to be fairly resistant to a successful timing attack but suggest some avenues to explore.
- This weakness is overcome by triple DES and AES algorithms.

DES DESIGN CONTROVERSY

- Although DES standard is public was considerable controversy over design -in choice of 56-bit key (vs Lucifer 128-bit)
- Subsequent events and public analysis show in fact design was appropriate
- DES has become widely used, especially in financial applications



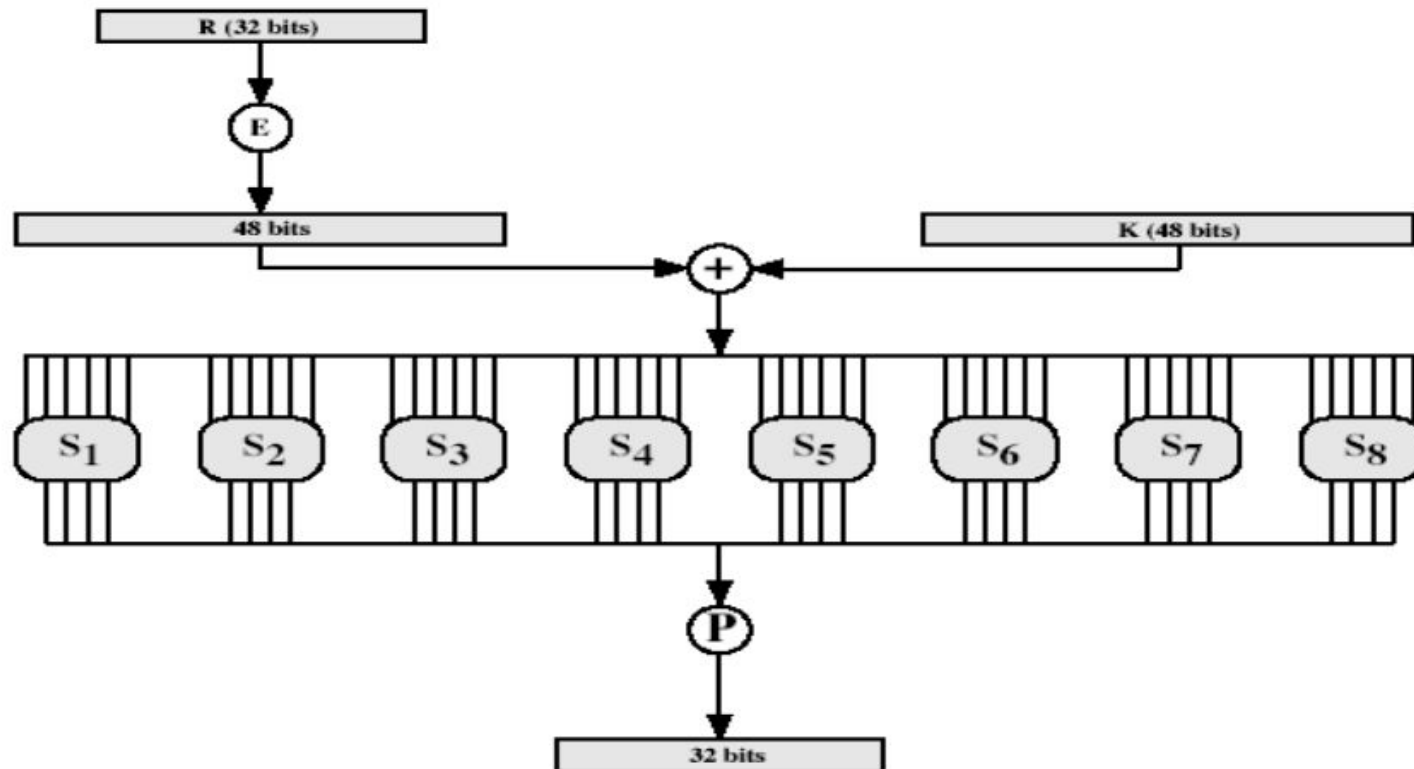
INITIAL PERMUTATION IP

- first step of the data computation IP reorders the input data bits quite regular in structure example:
- $IP(675a6967\ 5e5a6b5a) = (ffb2194d\ 004df6fb)$

DES ROUND STRUCTURE

- uses two 32-bit L & R halves
- as for any Feistel cipher can describe as: $L_i = R_{i-1}$ $R_i = L_{i-1} \text{ xor } F(R_{i-1}, K_i)$
- takes 32-bit R half and 48-bit subkey and:
 - expands R to 48-bits using Expansion Permutation E
 - adds to subkey
 - passes through 8 S-boxes to get 32-bit result
 - finally permutes this using 32-bit Permutation Function P

THE ROUND FUNCTION $F(R,K)$



SUBSTITUTION BOXES S

- 8 S-boxes
- Each S-Box maps 6 to 4 bits
 - – outer bits 1 & 6 (row bits) select the row
 - – inner bits 2-5 (col bits) select the column
 - – For example, in S1, for input 011001,
 - • the row is 01 (row 1)
 - • the column is 1100 (column 12)
 - • The value in row 1, column 12 is 9
 - • The output is 1001. • result is 8 X 4 bits, or 32 bits

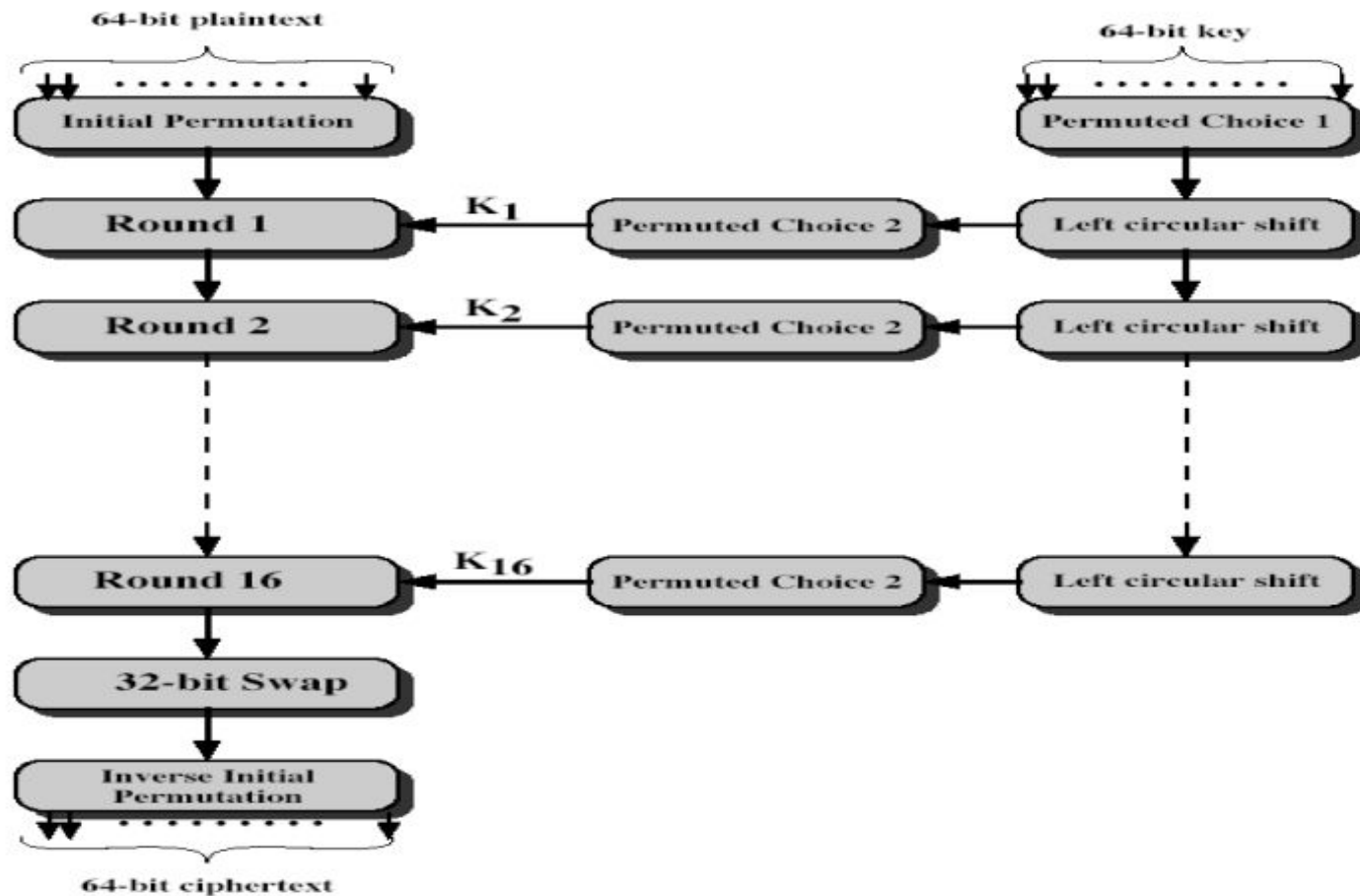
DES Key Schedule

- forms subkeys used in each round
- 1. initial permutation of the key PC1
- 2. divide the 56-bits in two 28-bit halves
- 3. at each round – 3.1. Left shift each half (28bits) separately either 1 or 2 places based on the left shift schedule
- Shifted values will be input for next round – 3.2. Combine two halves to 56 bits, permuting them by PC2 for use in function f
- PC2 takes 56-bit input, outputs 48 bits

DES DECRYPTION

- decrypt must unwind steps of data computation
- with Feistel design, do encryption steps again
- using subkeys in reverse order (SK16 ... SK1)
- note that IP undoes final FP step of encryption
- 1st round with SK16 undoes 16th encrypt round
- 16th round with SK1 undoes 1st encrypt round
- then final FP undoes initial encryption IP
- thus recovering original data value

DES Decryption (Reverse encryption)



AVALANCHE EFFECT

- key desirable property of encryption alg
- DES exhibits strong avalanche
- where a change of one input or key bit results in changing approx half
output bits

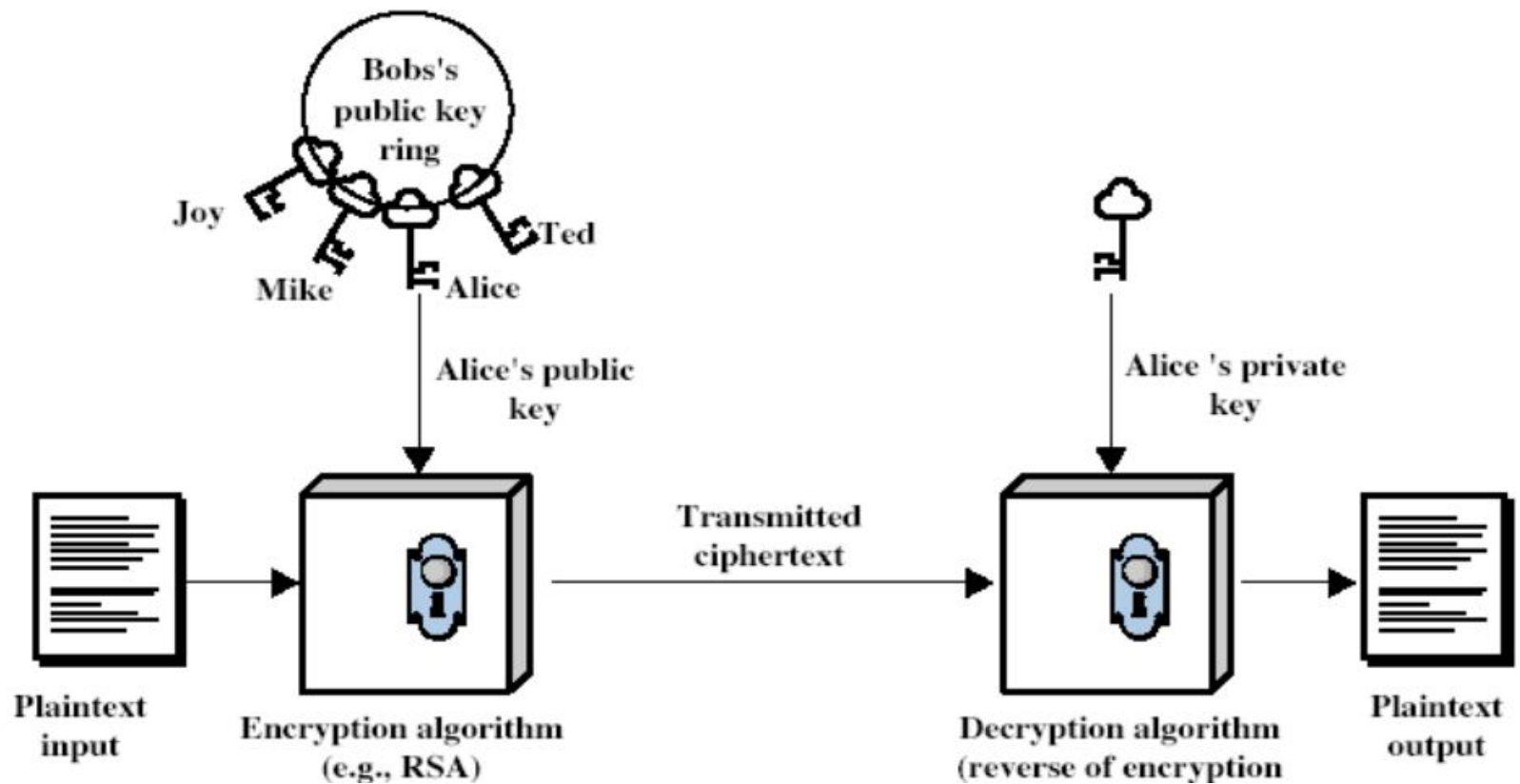
Principles of Public Cryptosystems

- Traditional private/secret/single-key cryptography uses one key shared by both sender and receiver.
- If this key is disclosed communications are compromised.
- Also in symmetric, parties are equal hence does not protect sender from receiver forging a message & claiming it's sent by sender
- Probably most significant advance in the 3000 year history of cryptography uses two keys – a public & a private key

Public-Key Cryptosystems

- Uses clever application of number theoretic concepts to function
- Complements rather than replaces private key crypto
- Public-key/two-key/asymmetric cryptography involves the use of two keys:
 - a public-key, which may be known by anybody, and can be used to encrypt messages, and verify signatures
 - a private-key, known only to the recipient, used to decrypt messages, and sign (create) signatures
- is asymmetric because – those who encrypt messages or verify signatures cannot decrypt messages or create signatures

Public-Key Cryptosystems



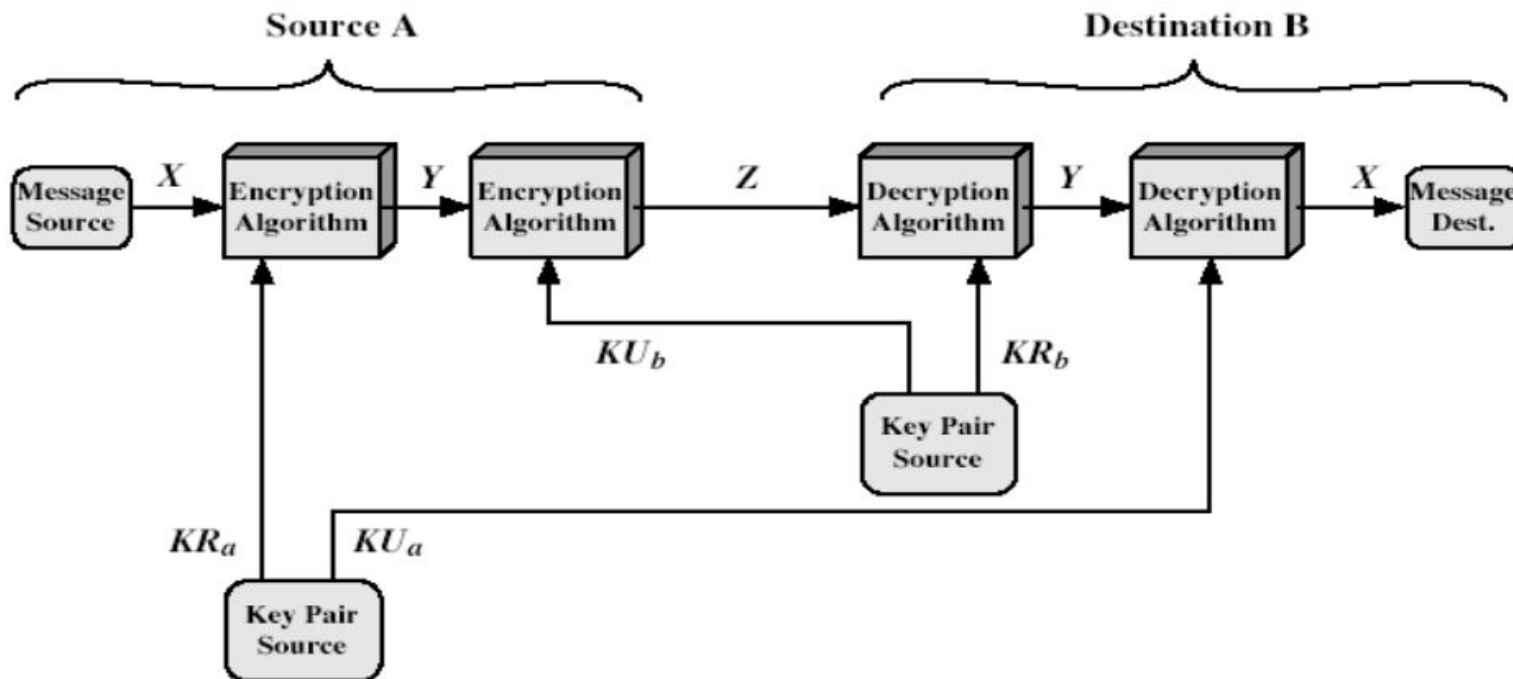
Applications for Public-Key Cryptosystems

- Developed to address two key issues:
 - – key distribution – how to have secure communications in general without having to trust a KDC with your key
 - – digital signatures – how to verify a message comes intact from the claimed sender

Requirements for Public-Key Cryptosystems

- Public-Key algorithms rely on two keys with the characteristics that it is:
 - – computationally infeasible to find decryption key knowing only algorithm & encryption key
 - – computationally easy to en/decrypt messages when the relevant (en/decrypt) key is known
 - – either of the two related keys can be used for encryption, with the other used for decryption (in some schemes)

Public-Key Cryptanalysis



PUBLIC-KEY APPLICATIONS

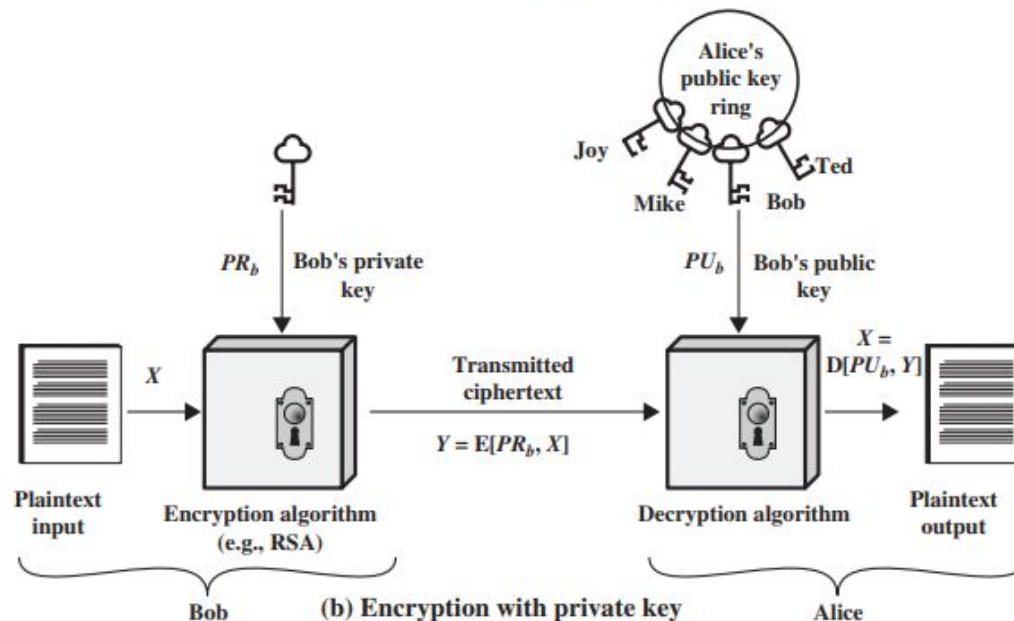
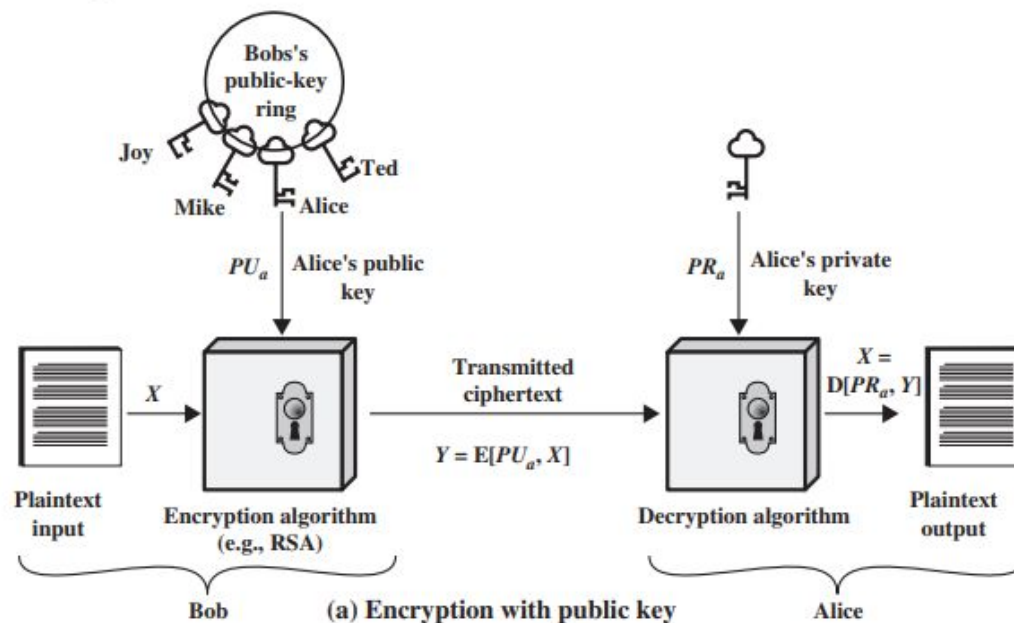
- Can classify uses into 3 categories:
 - –encryption/decryption (provide secrecy)
 - –digital signatures (provide authentication)
 - –key exchange (of session keys)
- some algorithms are suitable for all uses, others are specific to one

SECURITY OF PUBLIC KEY SCHEMES

- Like private key schemes brute force exhaustive search attack is always theoretically possible
- But keys used are too large (>512 bits)
- Security relies on a large enough difference in difficulty between easy (en/decrypt) and hard (cryptanalyse) problems
- More generally the hard problem is known, its just made too hard to do in practise
- Requires the use of very large numbers hence is slow compared to private key schemes

Public Key Cryptosystems

- Asymmetric algorithms rely on one key for encryption and a different but related key for decryption.
- Characteristics:
 - It is computationally infeasible to determine the decryption key given only knowledge of the cryptographic algorithm and the encryption key.
 - Either of the two related keys can be used for encryption, with the other used for decryption.



Public Key Cryptosystems

A public-key encryption scheme has these components:

1. Plaintext: This is the readable message or data that is fed into the algorithm as input.
2. Encryption algorithm: The encryption algorithm performs various transformations on the plaintext.
3. Public and private keys: This is a pair of keys that have been selected so that if one is used for encryption, the other is used for decryption.
4. Ciphertext: This is the scrambled message produced as output. It depends on the plaintext and the key. For a given message, two different keys will produce two different ciphertexts.
5. Decryption algorithm: This algorithm accepts the ciphertext and the matching key and produces the original plaintext.

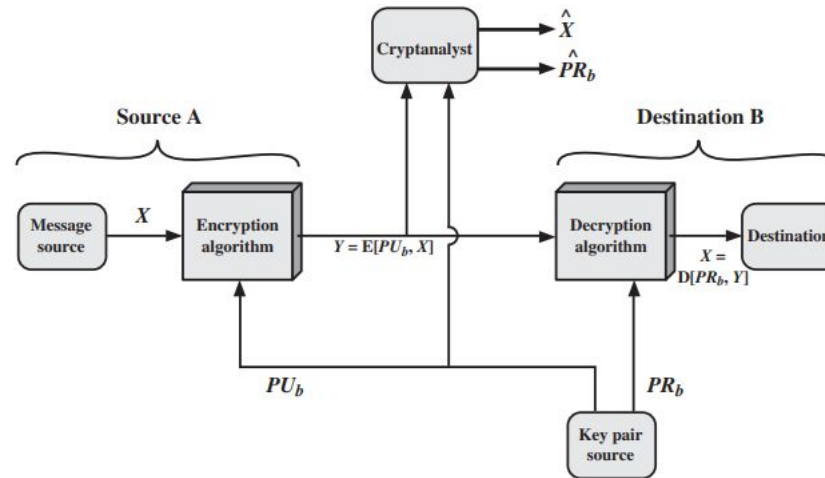
Private key vs Public Key Cryptosystems

Conventional Encryption	Public-Key Encryption
<i>Needed to Work:</i> 1. The same algorithm with the same key is used for encryption and decryption. 2. The sender and receiver must share the algorithm and the key. <i>Needed for Security:</i> 1. The key must be kept secret. 2. It must be impossible or at least impractical to decipher a message if the key is kept secret. 3. Knowledge of the algorithm plus samples of ciphertext must be insufficient to determine the key.	<i>Needed to Work:</i> 1. One algorithm is used for encryption and a related algorithm for decryption with a pair of keys, one for encryption and one for decryption. 2. The sender and receiver must each have one of the matched pair of keys (not the same one). <i>Needed for Security:</i> 1. One of the two keys must be kept secret. 2. It must be impossible or at least impractical to decipher a message if one of the keys is kept secret. 3. Knowledge of the algorithm plus one of the keys plus samples of ciphertext must be insufficient to determine the other key.

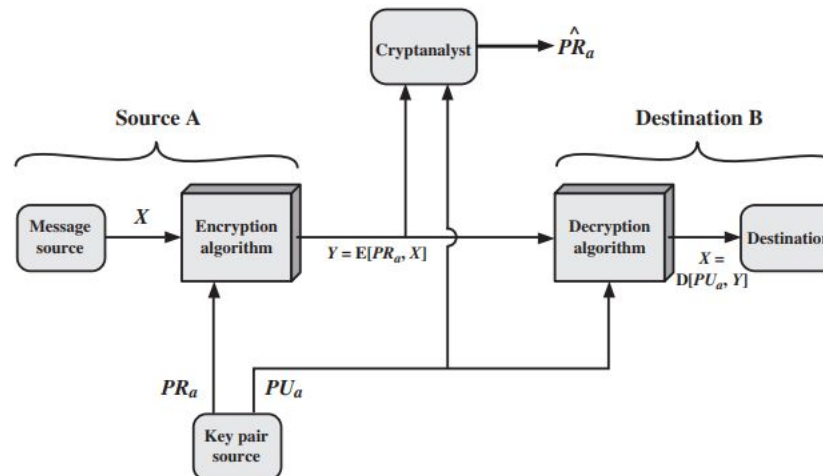
* Usually in symmetric key systems key is called as secret key and in asymmetric key systems the keys are called as private key and public key

Use of Asymmetric Cryptosystems:

- Confidentiality.



- Digital Signature.



Achieving confidentiality and authentication.

Confidentiality.

- When the message is encrypted using public key and decrypted using private key, the secrecy of the message is preserved.

$$Y = E(PU_b, X) \text{ and } X = D(PR_b, Y)$$

- Where PU and PR are public and private keys respectively.

Digital Signature.

- When message is encrypted using private key and decrypted using public key, authentication is achieved.

$$Y = E(PR_a, X) \text{ and } X = D(PU_a, Y)$$

Achieving both confidentiality and authentication.

- Usually a small block of bits that is a function of the document is encrypted using private key of the sender- authenticator.
- the authenticator has the property that it is infeasible to change the document without changing the authenticator.
- If the authenticator is encrypted with the sender's private key, it serves as a signature that verifies origin, content, and sequencing.
- The encryption process is represented as:

$$Z = E(PUb, E(PRa, X))$$

- The decryption process is represented as:

$$X = D(PUa, D(PRb, Z))$$

Achieving both confidentiality and authentication.

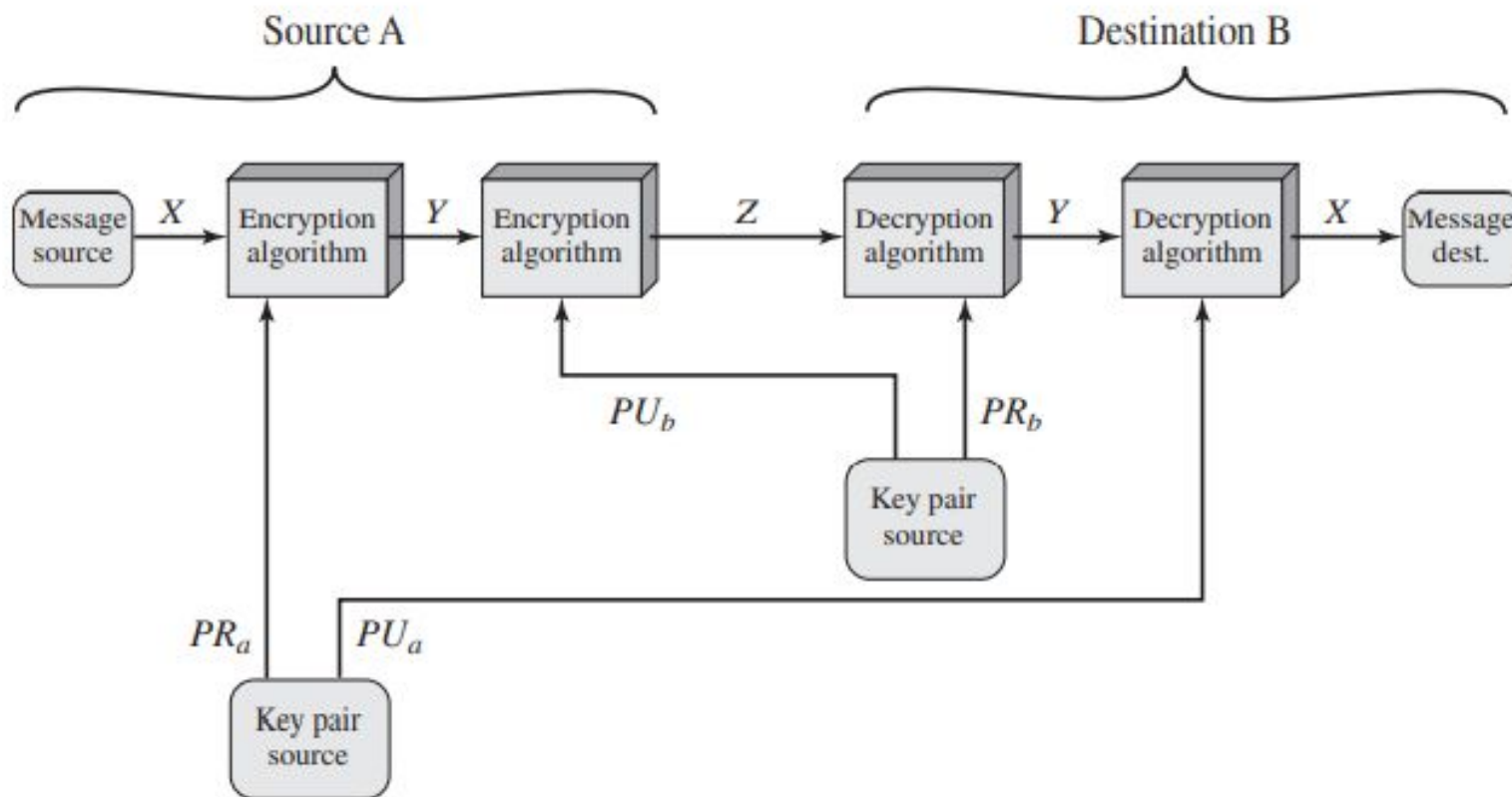


Figure 9.4 Public-Key Cryptosystem: Authentication and Secrecy

Applications for Public-Key Cryptosystems

Use of public-key cryptosystems :

1. Encryption/decryption: The sender encrypts a message with the recipient's public key.
2. Digital signature: The sender “signs” a message with its private key.

Signing is achieved by a cryptographic algorithm applied to the message or to a small block of data that is a function of the message.

3. Key exchange: Two sides cooperate to exchange a session key.

Requirements for Public-Key Cryptography

1. It is computationally easy for a party B to generate a pair (public key PU_b , private key PR_b).
2. It is computationally easy for a sender A, knowing the public key and the message to be encrypted, M , to generate the corresponding ciphertext: $C = E(PU_b, M)$
3. It is computationally easy for the receiver B to decrypt the resulting ciphertext using the private key to recover the original message: $M = D(PR_b, C) = D[PR_b, E(PU_b, M)]$
4. It is computationally infeasible for an adversary, knowing the public key, PU_b , to determine the private key, PR_b .
5. It is computationally infeasible for an adversary, knowing the public key, PU_b , and a ciphertext, C , to recover the original message, M .
6. The two keys can be applied in either order: $M = D[PU_b, E(PR_b, M)] = D[PR_b, E(PU_b, M)]$

The RSA Algorithm

1. The RSA scheme is a cipher in which the plaintext and ciphertext are integers between 0 and $n - 1$ for some n .
2. A typical size for n is 1024 bits, or 309 decimal digits. That is, n is less than 2^{1024} .
3. Plaintext is encrypted in blocks, with each block having a binary value less than some number n .
4. In practice, the block size is i bits, where $2^i < n \leq 2^{i+1}$.
5. Encryption and decryption are of the following form, for some plaintext block M and ciphertext block C .
6. $C = M^e \bmod n$ and $M = C^d \bmod n = (M^e)^d \bmod n = M^{ed} \bmod n$

The RSA Algorithm

1. Both sender and receiver must know the value of n .
2. The sender knows the value of e , and only the receiver knows the value of d .
3. Thus, this is a public-key encryption algorithm with a public key of $PU = \{e, n\}$ and a private key of $PR = \{d, n\}$.

RSA requirements:

1. It is possible to find values of e , d , and n such that $M^{ed} \bmod n = M$ for all $M < n$.
2. It is relatively easy to calculate $M^e \bmod n$ and $C^d \bmod n$ for all values of $M < n$.
3. It is infeasible to determine d given e and n .

The RSA Algorithm

$$M^{ed} \bmod n = M$$

- The above equation is true if e and d are multiplicative inverses modulo $\phi(n)$, where $\phi(n)$ is the Euler totient function.
- The relationship between e and d can be expressed as $ed \bmod \phi(n) = 1$.

Rewritten as:

$$ed = 1 \bmod \phi(n)$$

$$d = e^{-1} \bmod \phi(n)$$

Components of RSA:

- p, q , two prime numbers (private, chosen)
- $n = pq$ (public, calculated),
- e , with $\gcd(\phi(n), e) = 1$; $1 < e < \phi(n)$ (public, chosen)
- $d = e^{-1} \bmod \phi(n)$ (private, calculated)

The RSA Algorithm

- The private key consists of $\{d, n\}$ and the public key consists of $\{e, n\}$.
- Suppose that user A has published its public key and that user B wishes to send the message M to A.
- Then B calculates $C = M^e \bmod n$ and transmits C .
- On receipt of this ciphertext, user A decrypts by calculating $M = C^d \bmod n$.

Key Generation by Alice

Select p, q	p and q both prime, $p \neq q$
Calculate $n = p \times q$	
Calculate $\phi(n) = (p - 1)(q - 1)$	
Select integer e	$\gcd(\phi(n), e) = 1; 1 < e < \phi(n)$
Calculate d	$d \equiv e^{-1} \pmod{\phi(n)}$
Public key	$PU = \{e, n\}$
Private key	$PR = \{d, n\}$

Encryption by Bob with Alice's Public Key

Plaintext:	$M < n$
Ciphertext:	$C = M^e \bmod n$

Decryption by Alice with Alice's Public Key

Ciphertext:	C
Plaintext:	$M = C^d \bmod n$

The RSA Example:

1. Select two prime numbers, $p = 17$ and $q = 11$.
2. Calculate $n = pq = 17 \times 11 = 187$.
3. Calculate $\phi(n) = (p - 1)(q - 1) = 16 \times 10 = 160$.
4. Select e such that e is relatively prime to $\phi(n) = 160$ and less than $\phi(n)$; we choose $e = 7$.
5. Determine d such that $de \equiv 1 \pmod{160}$ and $d < 160$. The correct value is $d = 23$, because $23 \times 7 = 161 = (1 \times 160) + 1$; d can be calculated using the extended Euclid's algorithm (Chapter 4).

The resulting keys are public key $PU = \{7, 187\}$ and private key $PR = \{23, 187\}$. The example shows the use of these keys for a plaintext input of $M = 88$. For encryption, we need to calculate $C = 88^7 \bmod 187$. Exploiting the properties of modular arithmetic, we can do this as follows.

$$88^7 \bmod 187 = [(88^4 \bmod 187) \times (88^2 \bmod 187) \times (88^1 \bmod 187)] \bmod 187$$

$$88^1 \bmod 187 = 88$$

$$88^2 \bmod 187 = 7744 \bmod 187 = 77$$

$$88^4 \bmod 187 = 59,969,536 \bmod 187 = 132$$

$$88^7 \bmod 187 = (88 \times 77 \times 132) \bmod 187 = 894,432 \bmod 187 = 11$$

For decryption, we calculate $M = 11^{23} \bmod 187$:

$$11^{23} \bmod 187 = [(11^1 \bmod 187) \times (11^2 \bmod 187) \times (11^4 \bmod 187) \times (11^8 \bmod 187) \times (11^8 \bmod 187)] \bmod 187$$

$$11^1 \bmod 187 = 11$$

$$11^2 \bmod 187 = 121$$

$$11^4 \bmod 187 = 14,641 \bmod 187 = 55$$

$$11^8 \bmod 187 = 214,358,881 \bmod 187 = 33$$

$$11^{23} \bmod 187 = (11 \times 121 \times 55 \times 33 \times 33) \bmod 187 = 79,720,245 \bmod 187 = 88$$

The RSA Example:

- Generate two large random primes, p and q , of approximately equal size such that their product $n=pq$ is of the required bit length, e.g. 1024 bits.
- Compute $n=pq$ and $\phi=(p-1)(q-1)$.
- Choose an integer e , $1<e<\phi$, such that $\gcd(e,\phi)=1$.
- Compute the secret exponent d , $1<d<\phi$, such that $ed\equiv 1\pmod{\phi}$.
- The public key is (n,e) and the private key (d,p,q) .
- Keep all the values d , p , q and ϕ secret.
- ✓ n is known as the modulus.
- ✓ e is known as the public exponent or encryption exponent or just the exponent.
- ✓ d is known as the secret exponent or decryption exponent.

The RSA Example:

- $P=11$ and $q=3$
- $N=33$
- $\Phi=20$.
- Determine e such that $\gcd(\Phi, e)=1$. For this determine the factors of 20
- Factors are $= 2, 4, 5..$
- So e can be 3.

$$D * E \bmod (\Phi - 1) * (\Phi - 1) = 1.$$

$$(D * 3) \bmod 20 = 1$$

$$d = 7.$$

$$\text{Public key} = (n, e) = (33, 3)$$

$$\text{Private key} = (n, d) = (33, 7).$$

Now say we want to encrypt the message $m = 7$,

$$c = m^e \bmod n = 7^3 \bmod 33 = 343 \bmod 33 = 13$$

$c = m^e \bmod n = 7^3 \bmod 33 = 343 \bmod 33 = 13$. Hence the ciphertext $c = 13$.

To check decryption we compute

$$m' = c^d \bmod n = 13^7 \bmod 33 = 7$$

$m' = c^d \bmod n = 13^7 \bmod 33 = 7$.

The RSA Example:

- $P=11$ and $q=3$
- $N=33$
- $\Phi=20$.
- Determine e such that $\gcd(p(n),e)=1$. For this determine the factors of 20
- Factors are= 2,4,5..
- So e can be 3.

$$D * E \bmod (A - 1) * (B - 1) = 1.$$

$$(D * 3) \bmod 20 = 1$$

$$d = 7.$$

$$\text{Public key} = (n, e) = (33, 3)$$

$$\text{Private key} = (n, d) = (33, 7).$$

Now say we want to encrypt the message $m = 7$,

$$c = m_e \bmod n = 7_3 \bmod 33 = 343 \bmod 33 = 13 \quad c = m \bmod n = 73 \bmod 33 = 343 \bmod 33 = 13. \text{ Hence the ciphertext } c = 13.$$

To check decryption we compute

$$m' = c_d \bmod n = 13_7 \bmod 33 = 7 \quad m' = c \bmod n = 137 \bmod 33 = 7.$$

The RSA Example:

Now if we calculate the ciphertext c for all the possible values of m (0 to 32), we get

m	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
c	0	1	8	27	31	26	18	13	17	3	10	11	12	19	5	9	4
m	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	
c	29	24	28	14	21	22	23	30	16	20	15	7	2	6	25	32	

If we wanted to use this system to keep secrets, we could let $A=2$, $B=3$, ..., $Z=27$. (We specifically avoid 0 and 1 here for the reason given above). Thus the plaintext message "HELLOWORLD" would be represented by the set of integers $m_1, m_2, \dots$

(9, 6, 13, 13, 16, 24, 16, 19, 13, 5)

Using our table above, we obtain ciphertext integers $c_1, c_2, \dots$

(3, 18, 19, 19, 4, 30, 4, 28, 19, 26)

The RSA Example:

Now if we calculate the ciphertext c for all the possible values of m (0 to 32), we get

m	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
c	0	1	8	27	31	26	18	13	17	3	10	11	12	19	5	9	4
m	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	
c	29	24	28	14	21	22	23	30	16	20	15	7	2	6	25	32	

If we wanted to use this system to keep secrets, we could let $A=2$, $B=3$, ..., $Z=27$. (We specifically avoid 0 and 1 here for the reason given above). Thus the plaintext message "HELLOWORLD" would be represented by the set of integers $m_1, m_2, \dots$

(9, 6, 13, 13, 16, 24, 16, 19, 13, 5)

Using our table above, we obtain ciphertext integers $c_1, c_2, \dots$

(3, 18, 19, 19, 4, 30, 4, 28, 19, 26)

Diffie-Hellman Key Exchange.

*Go, change the
world*



Alice



Bob

Alice and Bob share a prime number q and an integer α , such that $\alpha < q$ and α is a primitive root of q

Alice generates a private key X_A such that $X_A < q$

Alice calculates a public key $Y_A = \alpha^{X_A} \bmod q$

Alice receives Bob's public key Y_B in plaintext

Alice calculates shared secret key $K = (Y_B)^{X_A} \bmod q$



Alice and Bob share a prime number q and an integer α , such that $\alpha < q$ and α is a primitive root of q

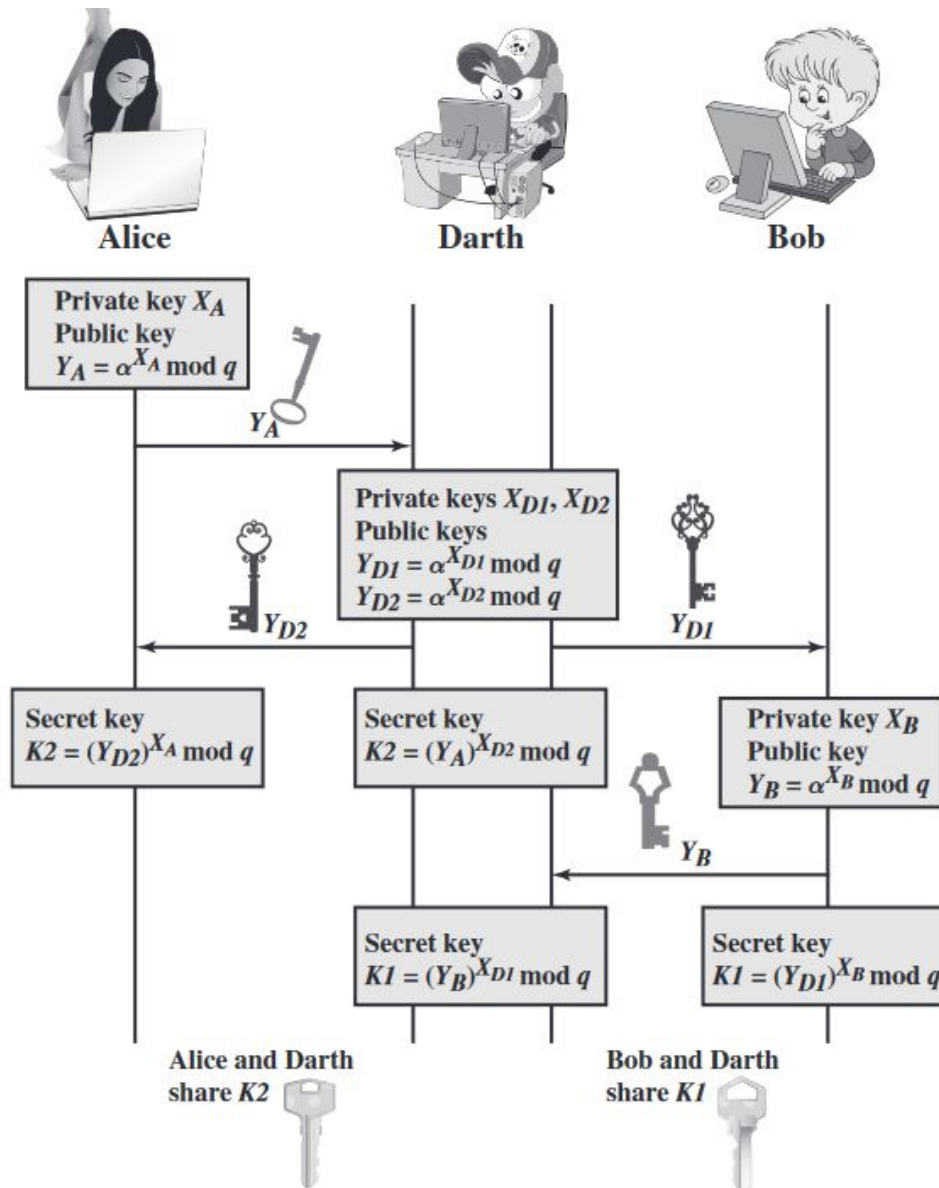
Bob generates a private key X_B such that $X_B < q$

Bob calculates a public key $Y_B = \alpha^{X_B} \bmod q$

Bob receives Alice's public key Y_A in plaintext

Bob calculates shared secret key $K = (Y_A)^{X_B} \bmod q$





1. Darth prepares for the attack by generating two random private keys X_{D1} and X_{D2} and then computing the corresponding public keys Y_{D1} and Y_{D2} .
2. Alice transmits Y_A to Bob.
3. Darth intercepts Y_A and transmits Y_{D1} to Bob. Darth also calculates $K2 = (Y_A)^{X_{D2}} \bmod q$.
4. Bob receives Y_{D1} and calculates $K1 = (Y_{D1})^{X_B} \bmod q$.
5. Bob transmits Y_B to Alice.
6. Darth intercepts Y_B and transmits Y_{D2} to Alice. Darth calculates $K1 = (Y_B)^{X_{D1}} \bmod q$.
7. Alice receives Y_{D2} and calculates $K2 = (Y_{D2})^{X_A} \bmod q$.

At this point, Bob and Alice think that they share a secret key, but instead Bob and Darth share secret key $K1$ and Alice and Darth share secret key $K2$. All future communication between Bob and Alice is compromised in the following way.

1. Alice sends an encrypted message M : $E(K2, M)$.
2. Darth intercepts the encrypted message and decrypts it to recover M .
3. Darth sends Bob $E(K1, M)$ or $E(K1, M')$, where M' is any message. In the first case, Darth simply wants to eavesdrop on the communication without altering it. In the second case, Darth wants to modify the message going to Bob.

DH does not authenticate the participants, using digital signature and PKI could prevent

Diffie-Hellman Algorithm

Example: Let $p=11$, $g=2$, then

a	1	2	3	4	5	6	7	8	9	10	11
g^a	2	4	8	16	32	64	128	256	512	1024	2048
$g^a \bmod p$	2	4	8	5	10	9	7	3	6	1	2

A chooses 4, B chooses 3, then shared secret is $(2^3)^4 = (2^4)^3 = 2^{12} = 4 \pmod{11}$

Adversaries sees $2^3=8$ and $2^4=5$, needs to solve one of $2^x=8$ and $2^y=5$ to figure out the shared secret.